Lectures on Monte Carlo Theory

Chapter 4: Simulation Output Analysis: Independent Replications

Based on Lorek & Rolski

Chapter Outline

Introduction: Why Output Analysis? I

Every time you run a Monte Carlo simulation, each result you get is a *random number* — it will be slightly different if you run the simulation again. Suppose you estimate $I = \pi$ (pi, approximately 3.14159) by simulating $R = 1{,}000$ random points and obtain the estimate $\hat{I} = 3.14$. That single number tells you almost nothing on its own. How close is 3.14 to the truth? Could the true value be 3.10 or 3.20 instead? How many more runs would you need to be confident your answer is correct to two decimal places?

Output analysis gives scientific, quantitative meaning to simulation results. Without it, a simulation study produces numbers but not knowledge. With it, we can make rigorous statements like: "I am 95% confident that the true value lies between 3.12 and 3.16."

The two central goals of output analysis in this chapter are:

1. Build a *confidence interval* (CI) that brackets the true value I with a specified probability $1 - \alpha$ (one minus alpha), for example 95% when $\alpha = 0.05$.
2. Determine how many replications R are needed to achieve a desired half-width b (the "radius" of the interval around the estimate) for that confidence level.

Introduction: Why Output Analysis? II

The mathematical foundation is the **Central Limit Theorem (CLT)**, which guarantees that the average of many independent random variables is approximately normally distributed regardless of the individual distribution. This allows us to use well-known properties of the normal distribution to build confidence intervals.

Introduction: Why Output Analysis? III

The Central Goal of Chapter 4

Everything in this chapter flows from the **fundamental relationship**:

$$b \;=\; \frac{z_{1-\alpha/2}\, D}{\sqrt{R}}.$$

Reading each symbol in plain English: b is the CI half-width (how precise the estimate is — smaller is better); R is the number of simulation replications (how many independent runs we perform); $z_{1-\alpha/2}$ (z sub one-minus-alpha-over-two) is the standard normal quantile at level $1-\alpha/2$, which for 95% confidence equals $z_{0.975} = 1.96$, and for 99% confidence equals $z_{0.995} = 2.576$; and D is a variability constant that depends on the specific estimation problem (it captures how "spread out" the simulation outputs are). Any two of these three quantities (b, R, D) determine the third.

Introduction: Why Output Analysis? IV

How to Read This Formula

The formula says that the error b shrinks as $1/\sqrt{R}$ — the square root of the number of replications. This means that to halve the error you must *quadruple* the number of runs. Going from $b = 0.1$ to $b = 0.01$ (ten times more precise) requires *one hundred times* as many replications. This "diminishing returns" property is a fundamental fact about Monte Carlo estimation that every practitioner must understand.

Introduction: Why Output Analysis? V

General Problem Setup

Throughout Chapter 4, we wish to estimate a quantity that can be written as an expectation:

$$I = \mathbb{E}Y,$$

where $\mathbb{E}$ (blackboard-bold E) denotes the *expectation* (the theoretical long-run average, also called the mean), and Y is a random variable that we can simulate on a computer. We require that Y has finite variance: $\mathbb{E}Y^2 < \infty$ (this rules out pathological distributions with infinite spread). We draw R independent, identically distributed (i.i.d.) copies $Y_1, Y_2, \ldots, Y_R$ — each one being a fresh, independent simulation run — and form their sample mean $\hat{Y}_R$ to estimate I.

Two fundamentally different notions of error will appear throughout this chapter, and they behave very differently.

Absolute error: The absolute error is $|\hat{I} - I|$, the raw numerical distance between the estimate and the truth. We want this to be at most some tolerance ε (epsilon) with high probability. For estimating $I = \pi$, we might ask for absolute error at most 0.01, meaning the estimate lies within the interval $(3.131, 3.151)$.

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Relative error: The relative error is $|\hat{I}/I - 1| = |\hat{I} - I|/I$, the discrepancy expressed as a fraction of the true value. A relative error of 1% means the estimate is within 1% of the truth. Relative error is much more informative when I itself is very small — for example, when I is the probability of a rare event like $I = 10^{-6}$. An absolute error of 0.01 would be completely useless there (it would allow the estimate to be anywhere from 0 to 0.0100001), whereas a relative error of 1% would mean the estimate is between 0.99×10^{-6} and 1.01×10^{-6} — genuinely informative.

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Error Control Conditions (for $\varepsilon, \delta \in (0,1)$)

We want our estimator $\hat{I}$ to satisfy one of the following with probability at least $1-\delta$ (delta, the allowed failure probability; for 99% confidence, $\delta = 0.01$):

$$\text{(E1) Absolute error: } \mathbb{P}\Big(|\hat{I} - I| \geq \varepsilon\Big) \leq \delta,$$

$$\text{(E2) Relative error: } \mathbb{P}\Big(|\hat{I}/I - 1| \geq \varepsilon\Big) \leq \delta.$$

Condition (E2) is strictly harder than (E1) whenever $I < 1$, because it demands that the error be small *relative to the true value*, not just in absolute terms.

Introduction: Why Output Analysis? VIII

Key Takeaway

This chapter focuses primarily on absolute error control for Sections 4.1 through 4.3, building up the machinery of confidence intervals for means, transformed means, and quantiles. Section 4.4 then returns to relative error, showing it is dramatically harder when I is small — which motivates the variance reduction techniques of Chapter 5.

4.1.1 The Sample Mean Estimator I

We want to estimate $I = \mathbb{E}Y$. The most natural estimator is simply the average of R independent simulation outputs — one output per run. This estimator is so fundamental that it is worth studying in careful detail, because all the ideas carry over to more complex situations.

Sample Mean Estimator

Given R independent replications $Y_1, Y_2, \ldots, Y_R$ (each one is an independent simulation run with the same distribution as Y), the sample mean estimator is:

$$\hat{Y}_R \; = \; \frac{1}{R} \sum_{j=1}^{R} Y_j.$$

Here $\sum_{j=1}^{R}$ (capital Sigma, meaning "sum") adds up the simulation outputs from run number $j = 1$ to run number $j = R$. The symbol R (capital R) is the total number of replications, and Y_j (Y sub j) is the output of the j-th run.

4.1.1 The Sample Mean Estimator II

This estimator has two fundamental properties, which together guarantee that it is a trustworthy way to estimate I.

Unbiasedness. On average, the sample mean equals the truth. Taking the expectation of both sides of the formula:

$$\mathbb{E}\hat{Y}_R = \mathbb{E}\left[\frac{1}{R}\sum_{j=1}^{R} Y_j\right] = \frac{1}{R}\sum_{j=1}^{R} \mathbb{E}Y_j = \frac{1}{R} \cdot R \cdot \mathbb{E}Y = I.$$

Since each Y_j has the same distribution as Y, we have $\mathbb{E}Y_j = I$ for each j. The result $\mathbb{E}\hat{Y}_R = I$ means the estimator does not systematically over- or under-estimate — it does not "lean" in any direction.

Strong consistency. By the Strong Law of Large Numbers (SLLN), as we run more and more replications, the sample mean converges to the true value with probability 1:

$$\hat{Y}_R \;\longrightarrow\; I \quad \text{with probability 1, as } R \to \infty.$$

In plain terms: if we could run infinitely many simulations, the average would be exactly right. For any fixed gap $b > 0$, this implies *weak consistency*: the probability of the estimate being wrong by more than b vanishes as R grows: $\mathbb{P}(|\hat{Y}_R - I| \geq b) \to 0$.

4.1.1 The Sample Mean Estimator III

Weak consistency in turn guarantees that for any desired accuracy $b > 0$ and any failure probability $\alpha \in (0, 1)$, there exists a sample size R_0 such that for all $R \geq R_0$:

$$\mathbb{P}\Big(\hat{Y}_R - b < I \leq \hat{Y}_R + b\Big) \geq 1 - \alpha. \tag{1.2}$$

This is the key promise: for large enough R, the random interval $(\hat{Y}_R - b,\ \hat{Y}_R + b)$ brackets the true I with high probability.

4.1.1 The Sample Mean Estimator IV

The guarantee in (1.2) is existential — it tells us such R_0 exists but gives no formula for how large it is. Two tools give us quantitative, computable bounds.

4.1.1 The Sample Mean Estimator V

Chebyshev's Inequality-Based Bound (Remark 4.1.1)

Chebyshev's inequality states that for any random variable X with finite variance and any $t > 0$:

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \;\leq\; \frac{\mathrm{Var}(X)}{t^2}.$$

Here $\mathrm{Var}(X)$ is the variance of X (how spread out X is around its mean), and t is the threshold distance. Since $\hat{Y}_R$ has variance $\mathrm{Var}(\hat{Y}_R) = \sigma_Y^2/R$ (where $\sigma_Y^2 = \mathrm{Var}(Y)$ is the variance of a single observation), applying Chebyshev with $t = b$:

$$\mathbb{P}(|\hat{Y}_R - I| \geq b) \;\leq\; \frac{\sigma_Y^2/R}{b^2}.$$

Setting this bound equal to α (alpha, the significance level, e.g. $\alpha = 0.05$ for 95% confidence) and solving for the half-width b gives the *Chebyshev half-width*:

$$b_R^{\mathrm{Ch}} \;=\; \frac{\sigma_Y}{\alpha^{1/2}\sqrt{R}}. \tag{1.3}$$

Here $\sigma_Y = \sqrt{\mathrm{Var}(Y)}$ (sigma sub Y) is the standard deviation of a single simulation output — how

4.1.1 Confidence Intervals in Practice I

In practice, the standard deviation $\sigma_Y = \sqrt{\mathrm{Var}(Y)}$ is unknown — we do not know the distribution of Y a priori (if we did, we would not need to simulate). We must estimate σ_Y^2 from the same simulation data we use to estimate I.

4.1.1 Confidence Intervals in Practice II

Sample Variance (Two Definitions)

Given observations $Y_1, \ldots, Y_R$ with sample mean $\hat{Y}_R$, we have two common estimators of the variance $\sigma_Y^2 = \mathrm{Var}(Y)$:

$$\hat{S}_R^2 = \frac{1}{R-1} \sum_{j=1}^{R} (Y_j - \hat{Y}_R)^2 \quad \text{(unbiased: } \mathbb{E}\hat{S}_R^2 = \sigma_Y^2\text{)},$$

$$\hat{\sigma}_R^2 = \frac{1}{R} \sum_{j=1}^{R} (Y_j - \hat{Y}_R)^2 \quad \text{(biased, but slightly simpler).}$$

In both formulas, $(Y_j - \hat{Y}_R)^2$ (Y sub j minus Y-hat sub R, squared) is the squared deviation of the j-th observation from the sample mean. Summing these squared deviations and dividing gives the average squared deviation — the sample variance.
The divisor $R-1$ in $\hat{S}_R^2$ is called *Bessel's correction*. It compensates for the fact that we computed the deviations around $\hat{Y}_R$ (which was itself estimated from the data) rather than around the true mean I.
For large R, both definitions are nearly identical, but $\hat{S}_R^2$ is preferred for its exact unbiasedness.

4.1.1 Examples: Estimating π (pi) I

These two examples both estimate $I = \pi \approx 3.14159$, yet they define different random variables Y with $\mathbb{E}Y = \pi$. The variance of Y directly controls how many replications are needed — so choosing a better Y directly saves computation, even before we apply any sophisticated variance reduction techniques.

4.1.1 Examples: Estimating π (pi) II

Example 4.1.6 (Asmussen & Glynn) — Hit-or-Miss Estimator

Imagine throwing a dart uniformly at random at the unit square $[0, 1]^2$. The dart falls inside the quarter-circle of radius 1 (centered at the origin, covering the quarter of the unit disk in the first quadrant) if and only if $U_1^2 + U_2^2 \leq 1$, which happens with probability $\pi/4$ (the area of the quarter-circle).
We define the random variable $Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$, where $\mathbf{1}(\cdot)$ (indicator function) equals 1 if the condition is true and 0 otherwise, and (U_1, U_2) are independent uniform random variables on $[0, 1]$.
Then:

$$\mathbb{E}Y = 4 \cdot \mathbb{P}(U_1^2 + U_2^2 \leq 1) = 4 \cdot \frac{\pi}{4} = \pi.$$

Since $\mathbf{1}(U_1^2 + U_2^2 \leq 1)$ is a Bernoulli random variable with parameter $p = \pi/4$, we have $\mathrm{Var}(\mathbf{1}(\cdot)) = p(1 - p) = (\pi/4)(1 - \pi/4)$, so:

$$\mathrm{Var}(Y) = 16 \cdot \frac{\pi}{4}\left(1 - \frac{\pi}{4}\right) = 16 \times 0.7854 \times 0.2146 \approx 2.697.$$

For second-digit accuracy ($b = 0.05$), formula (1.7) gives: $R \approx 1.96^2 \times 2.697/0.05^2 \approx 4{,}144$ replications.

4.1.1 Crude Monte Carlo and Hit-or-Miss Integration I

A core application of Monte Carlo is computing integrals. Suppose we want to evaluate:

$$I = \int_B k(x)\,dx,$$

where $k : B \to \mathbb{R}$ is an integrable function (we can compute $k(x)$ for any given x) and $B \subseteq \mathbb{R}^d$ is a bounded region with volume $\mathrm{Vol}(B) = 1$ (we can always rescale so that the volume equals 1; if $B = [0,1]^d$, it is already the case). We cannot compute this integral analytically, but we can evaluate k at any point.

Crude Monte Carlo (CMC) Estimator (Eq. 1.13)

Generate $V_1, \ldots, V_R$ uniformly and independently from B (written $V_j \sim \mathcal{U}(B)$). Set $Y = k(V)$, so that:

$$\mathbb{E}Y = \mathbb{E}[k(V)] = \int_B k(v) \frac{1}{\mathrm{Vol}(B)} \, dv = \int_B k(v) \, dv = I.$$

The CMC estimator averages the function values at these random points:

$$\hat{Y}_R^{\mathrm{CMC}} = \frac{1}{R} \sum_{j=1}^{R} k(V_j). \tag{1.13}$$

More generally, if X has density f on B and $Y = k(X)/f(X)$, then $\mathbb{E}Y = \int_B k(x) \, dx = I$, allowing us to use any convenient sampling distribution f.

4.1.1 Crude Monte Carlo and Hit-or-Miss Integration III

Example 4.1.10 (Madras) — Estimating $I = \int_0^1 k(x)\,dx$

Let $k(x) = (e^x - 1)/(e-1)$, where $e \approx 2.71828$ is Euler's number. Setting $Y = k(U)$ with $U \sim \mathcal{U}[0,1]$:

$$I = \mathbb{E}Y = \int_0^1 \frac{e^x - 1}{e-1}\,dx = \frac{1}{e-1}[e^x - x]_0^1 = \frac{e-1-1}{e-1} = \frac{e-2}{e-1} \approx 0.418.$$

$$\mathbb{E}Y^2 = \int_0^1 k^2(x)\,dx \approx 0.2567.$$

$$\sigma_Y^2 = \mathbb{E}Y^2 - (\mathbb{E}Y)^2 = 0.2567 - (0.418)^2 \approx 0.0820.$$

To achieve half-width b: $R = 1.96^2 \times 0.0820/b^2 \approx 0.328/b^2$. For $b = 0.01$ (two decimal places): $R \approx 3{,}280$ replications.

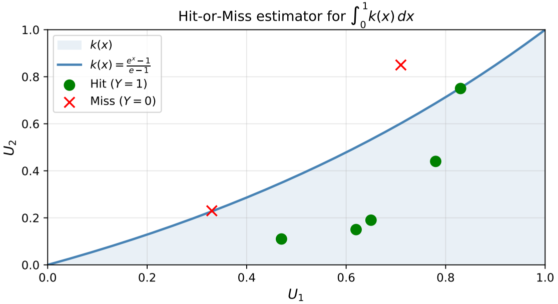
4.1.1 Crude Monte Carlo and Hit-or-Miss Integration IV

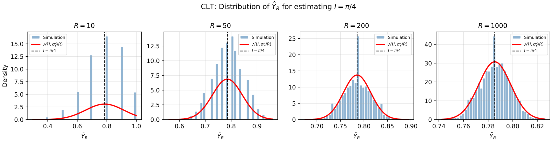
Hit-or-Miss (HoM) Estimator (Example 4.1.11)

Assume $0 \le k(x) \le 1$ for all $x \in [0,1]$. Generate two independent uniform random variables $U_1, U_2 \sim \mathcal{U}[0,1]$ and define $Y' = \mathbf{1}(k(U_1) > U_2)$ (the indicator that the point (U_1, U_2) falls below the curve $y = k(x)$). Then:

$$\mathbb{E}Y' = \int_0^1 \int_0^{k(u_1)} du_2 \, du_1 = \int_0^1 k(u_1) \, du_1 = I.$$

The HoM estimator averages these indicators: $\hat{Y}_R^{\mathrm{HoM}} = \frac{1}{R} \sum_{j=1}^{R} Y'_j$.
However, Y' is a Bernoulli random variable with parameter I, so $\mathrm{Var}(Y') = I(1-I)$. For the function $k(x) = (e^x - 1)/(e-1)$ with $I \approx 0.418$: $\mathrm{Var}(Y') \approx 0.418 \times 0.582 \approx 0.2433$. This is about $0.2433/0.0820 \approx 2.97\times$ the CMC variance. So HoM requires about 3 times as many replications as CMC for the same accuracy, and it also uses *two* uniform random numbers instead of one. The hit-or-miss method is simple to visualize but is statistically inefficient.


Hit-or-Miss estimator for $\int_0^1 k(x)\,dx$
$k(x)$
$k(x) = \frac{e^x - 1}{e - 1}$
Hit ($Y = 1$)
Miss ($Y = 0$)
U_2
U_1
0.0
0.2
0.4
0.6
0.8
1.0


CLT: Distribution of $\hat{Y}_R$ for estimating $I = \pi/4$
$R = 10$
$R = 50$
$R = 200$
$R = 1000$
Simulation
$\mathcal{N}(I, \sigma_Y^2/R)$
$I = \pi/4$
Density
$\hat{Y}_R$

4.1.1 Bounding the Variance Without Knowing I I

To plan a simulation study using formula (1.7), we need $\sigma_Y^2 = \mathrm{Var}(Y)$. But $\mathrm{Var}(Y) = \mathbb{E}Y^2 - (\mathbb{E}Y)^2 = \mathbb{E}Y^2 - I^2$ involves the unknown quantity I that we are trying to estimate! Sometimes we can bypass this circular problem by bounding σ_Y^2 from above using only known upper and lower bounds on the function k.

Variance Bound for CMC (Remark 4.1.12)

For the CMC estimator $Y = k(X)$ with $X \sim f$ on B, the variance is $\sigma_Y^2 = \int_B k^2(x)\, f(x)\, dx - I^2$. If we know constants k_{up} (an upper bound: $k(x) \le k_{\mathrm{up}}$ for all x) and k_{down} (a lower bound: $I \ge k_{\mathrm{down}}$), then:

$$\sigma_Y^2 \;=\; \int_B k^2(x)\, f(x)\, dx - I^2 \;\le\; k_{\mathrm{up}}^2 - k_{\mathrm{down}}^2.$$

Using this upper bound in formula (1.7) gives a conservative (slightly too large) required sample size R, but guarantees the desired confidence level without knowing I.

4.1.1 Bounding the Variance Without Knowing I II

Bounding $\mathrm{Var}(Y)$ for $k(x) = (e^x - 1)/(e - 1)$ on $[0, 1]$

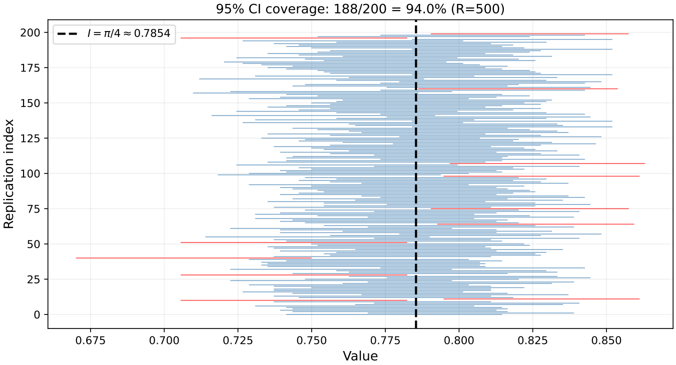
We know $k(0) = 0$ and $k(1) = 1$, so $0 \le k(x) \le 1$ for $x \in [0, 1]$. We also know $I = \mathbb{E}Y = (e - 2)/(e - 1) \approx 0.418 \ge 0$.

A more careful analysis using $e \approx 2.71828$ gives: $k_{\text{down}} \approx 0.4157$ and $\int_0^1 k^2(x)\, dx \le 0.2622$. Therefore:

$$\sigma_Y^2 \;\le\; 0.2622 - (0.4157)^2 \approx 0.0894.$$

The true value is $\sigma_Y^2 \approx 0.0820$, so the bound is about 9% too large. Using this bound, the required R is only 9% more than strictly necessary — a negligible price for not needing to know I in advance.

4.1.1 Bounding the Variance Without Knowing I III



4.1.1 Python: Computing a Confidence Interval

```
import numpy as np
from scipy import stats

def estimate_pi_ci(R, alpha=0.05, seed=42):
    """Estimate pi via hit-or-miss MC and report a (1-alpha) CI."""
    rng = np.random.default_rng(seed)
    U = rng.random((R, 2))
    # Y_j = 4 * 1(U1^2 + U2^2 <= 1); E[Y_j] = pi
    Y = 4.0 * (U[:, 0]**2 + U[:, 1]**2 <= 1).astype(float)

    Y_bar = Y.mean()              # sample mean -- estimator of pi
    S2    = Y.var(ddof=1)         # unbiased sample variance (ddof=1 => divide by R-1)
    S     = np.sqrt(S2)

    z      = stats.norm.ppf(1 - alpha / 2)   # z_{1-alpha/2}; for alpha=0.05 => 1.96
    margin = z * S / np.sqrt(R)              # half-width b

    return Y_bar, Y_bar - margin, Y_bar + margin

# Example: R=10000, alpha=0.05
Y_bar, lo, hi = estimate_pi_ci(R=10_000)
print(f"Estimate: {Y_bar:.4f}")
print(f"95% CI : ({lo:.4f}, {hi:.4f})")
print(f"True pi: {np.pi:.4f}")
# Typical output:
# Estimate: 3.1464
# 95% CI : (3.1142, 3.1786)
```

4.1.2 The Multidimensional Case I

Sometimes the quantity of interest is not a single number but a vector $\mathbf{I} = (I^{(1)}, \ldots, I^{(d)})^T = \mathbb{E}\mathbf{Y}$, where $\mathbf{Y} = (Y^{(1)}, \ldots, Y^{(d)})^T$ is a d-dimensional random vector (each component is a separate simulation output of interest). The superscript T denotes the transpose (turning a row into a column). For instance, we might simultaneously estimate the mean weight $I^{(1)} = \mathbb{E}[\text{weight}]$ and mean height $I^{(2)} = \mathbb{E}[\text{height}]$ of a population ($d = 2$), or compute d different integrals in a single simulation campaign.

Vector Sample Mean Estimator

Given i.i.d. replications $\mathbf{Y}_1, \ldots, \mathbf{Y}_R$ (each $\mathbf{Y}_j$ is a d-vector of simulation outputs from run j):

$$\hat{\mathbf{Y}}_R = \frac{1}{R}\sum_{j=1}^{R} \mathbf{Y}_j = \left(\hat{Y}_R^{(1)}, \ldots, \hat{Y}_R^{(d)}\right)^T,$$

where each component $\hat{Y}_R^{(k)}$ (Y-hat sub R superscript k) is the sample mean of the k-th coordinate across all R runs. This estimator is unbiased: $\mathbb{E}\hat{\mathbf{Y}}_R = \mathbf{I}$.

Let $\boldsymbol{\Sigma}_{\mathbf{Y}} = \mathbb{E}[(\mathbf{Y} - \mathbf{I})(\mathbf{Y} - \mathbf{I})^T]$ (bold Sigma sub Y) denote the $d \times d$ *covariance matrix* of $\mathbf{Y}$. Its (i, j)-entry is $\boldsymbol{\Sigma}_{\mathbf{Y}}(i, j) = \mathrm{Cov}(Y^{(i)}, Y^{(j)})$ — the covariance between the i-th and j-th components. The diagonal entries are the individual variances.

Multivariate Central Limit Theorem (Eq. 1.23)

As $R \to \infty$:

$$\sqrt{R}\,(\hat{\mathbf{Y}}_R - \mathbf{I}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathbf{Y}})\,.$$

This means that for large R, the vector estimator $\hat{\mathbf{Y}}_R$ is approximately distributed as a d-dimensional normal (Gaussian) with mean vector $\mathbf{I}$ and covariance matrix $\boldsymbol{\Sigma}_{\mathbf{Y}}/R$. The covariance matrix tells us both how uncertain each component is and how the uncertainties are correlated with each other.

Proposition 4.1.13 — Key Properties of the Multivariate Normal

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where $\boldsymbol{\mu}$ (bold mu) is the mean vector and $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^T$ (with $\mathbf{A}$ a square matrix, e.g. the Cholesky factor of $\boldsymbol{\Sigma}$). Then:

- (a) $\mathbf{Z} = \mathbf{A}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}(\mathbf{0}, \mathbf{Id}_d)$, where $\mathbf{Id}_d$ is the $d \times d$ identity matrix. In words: the standardized vector $\mathbf{Z}$ has independent standard normal components (no correlations).
- (b) $(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_d$ (chi-squared, pronounced "kai-squared", with d degrees of freedom).

Understanding part (b). The expression $(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$ is the *squared Mahalanobis distance* from $\mathbf{X}$ to its mean $\boldsymbol{\mu}$. It measures how far $\mathbf{X}$ is from its center in units of standard deviations, properly accounting for correlations between components. Part (b) says this distance follows a chi-squared (χ^2, chi-square) distribution with d degrees of freedom.

Proof sketch of (b). From (a), $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{Id}_d)$ has i.i.d. standard normal components $Z_1, \ldots, Z_d$. Then:

$$(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) = \mathbf{Z}^T \mathbf{Z} = \sum_{i=1}^{d} Z_i^2,$$

which is the sum of d squared independent standard normals. By definition, this sum has the χ^2_d distribution.

Critical values of the chi-squared distribution. To build 95% confidence regions, we need $q_{0.95}(\chi^2_d)$, the value below which 95% of the χ^2_d distribution falls:

Dimension d	$q_{0.95}(\chi^2_d)$	Interpretation
1	3.841	Equivalent to ± 1.96 in 1D (since $1.96^2 = 3.841$)
2	5.991	2D confidence ellipse
5	11.07	5D confidence ellipsoid
10	18.31	10D confidence ellipsoid

As d grows, the critical value grows, reflecting that a d-dimensional normal vector is increasingly likely to be far from the origin in at least one direction.

4.1.2.2 Shapes of Confidence Regions I

For $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, we want to find a region $\mathcal{C} \subset \mathbb{R}^d$ such that $\mathbb{P}(\mathbf{X} \in \mathcal{C}) \geq 1 - \alpha$. There are three natural shapes for such regions, each with different trade-offs between tightness and computational convenience.

(1) Confidence Ellipsoid — The Smallest Region (Eq. 1.15)

Using Proposition 4.1.13(b), the set:

$$\mathcal{C}^{\text{el}} = \left\{\mathbf{x} \in \mathbb{R}^d : \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \mathbf{x} \leq q_{1-\alpha}(\chi^2_d)\right\}$$

satisfies $\mathbb{P}(\mathbf{X} \in \mathcal{C}^{\text{el}}) = 1 - \alpha$ exactly. This is the minimum-volume region among all sets with coverage $1 - \alpha$.

If $\boldsymbol{\Sigma} = \mathbf{F}\boldsymbol{\Lambda}\mathbf{F}^T$ is the eigendecomposition (where $\boldsymbol{\Lambda}$ (Lambda) is the diagonal matrix of eigenvalues $\lambda_1, \ldots, \lambda_d$, and $\mathbf{F}$ is the matrix of corresponding eigenvectors $\mathbf{f}_1, \ldots, \mathbf{f}_d$), then the ellipsoid has principal axes of half-length $\sqrt{q_{1-\alpha}(\chi^2_d)\,\lambda_i}$ in direction $\mathbf{f}_i$. Directions where the variance (λ_i) is large produce longer axes.

4.1.2.2 Shapes of Confidence Regions II

(2) Confidence Parallelogram (Eq. 1.18)

Since $\boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu}) \sim \mathcal{N}(\mathbf{0}, \mathbf{Id}_d)$ (a standardized normal vector), we can build a hypercube (box) $\mathcal{H} = (-z, z)^d$ in the standardized space, where $z = \Phi^{-1}\big((\sqrt[d]{1-\alpha}+1)/2\big)$. Transforming back to the original space gives a parallelogram:

$$\mathcal{C}^{\mathrm{par}} = \boldsymbol{\Sigma}^{1/2}\mathcal{H} + \boldsymbol{\mu}.$$

This region has the same nominal coverage $1-\alpha$ but is generally larger than the ellipsoid (the ellipsoid is optimal). The advantage of the parallelogram is that it is aligned with the principal axes of the covariance matrix.

(3) Rectangular Confidence Region — Conservative and Simple (Eq. 1.19)

The axis-aligned rectangle:

$$\mathcal{C}^{\mathrm{rec}} = [-c_1, c_1] \times \cdots \times [-c_d, c_d], \quad c_i = \sigma_i \, \Phi^{-1}\left(\frac{\sqrt[d]{1-\alpha}+1}{2}\right),$$

where $\sigma_i = \sqrt{\mathbf{\Sigma}(i,i)}$ (sigma sub i) is the marginal standard deviation of coordinate i, satisfies $\mathbb{P}(\mathbf{X} \in \mathcal{C}^{\mathrm{rec}}) \geq 1 - \alpha$ by the Šidák inequality. This region is *conservative*: it generally achieves more than $1-\alpha$ coverage. Its advantage is that it is simply d independent confidence intervals — one per component — requiring no knowledge of the off-diagonal covariances.

4.1.2.2 Shapes of Confidence Regions IV

Example 4.1.14: Covariance Matrix $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$, $\alpha = 0.05$

The eigenvalues of $\boldsymbol{\Sigma}$ are found by solving $\det(\boldsymbol{\Sigma} - \lambda \mathbf{Id}) = 0$:
$(1-\lambda)(6-\lambda) - 4 = 0 \Rightarrow \lambda^2 - 7\lambda + 2 = 0$, giving $\lambda_{1,2} = (7 \pm \sqrt{41})/2$. Numerically: $\lambda_1 \approx 6.7015$ and $\lambda_2 \approx 0.2984$.
The corresponding eigenvectors (principal axes of the ellipsoid) are: $\mathbf{f}_1 \approx (-0.3310, -0.9436)^T$ and $\mathbf{f}_2 \approx (-0.9436, 0.3310)^T$.
At $\alpha = 0.05$: $q_{0.95}(\chi^2_2) = 5.991$.
Ellipsoid: Principal axis half-lengths are $\sqrt{5.991 \times 6.7015} \approx 6.337$ (in direction $\mathbf{f}_1$) and $\sqrt{5.991 \times 0.2984} \approx 1.337$ (in direction $\mathbf{f}_2$). The ellipsoid is highly elongated: about 5 times longer in the $\mathbf{f}_1$ direction.
Parallelogram: The threshold is $z = \Phi^{-1}((\sqrt{0.95}+1)/2) = 2.2365$. The corners are $(\pm 2.2365, \pm 2.2365)$ in standardized space, mapped back through $\boldsymbol{\Sigma}^{1/2}$.
Rectangle: $c_1 = 1 \times 2.2365 = 2.2365$ (for the first coordinate with $\sigma_1 = \sqrt{1} = 1$) and $c_2 = \sqrt{6} \times 2.2365 \approx 5.478$ (for the second coordinate with $\sigma_2 = \sqrt{6}$). The actual coverage of this rectangle is about 95.98% — slightly conservative, as expected.

4.1.2.3 Confidence Regions for the Estimator $\hat{\mathbf{Y}}_R$ I

In practice, we do not know $\mathbf{I}$ or $\mathbf{\Sigma}_{\mathbf{Y}}$ — these are exactly what we are trying to learn from the simulation. The multivariate CLT tells us that the estimator $\hat{\mathbf{Y}}_R$ is approximately $\mathcal{N}(\mathbf{I}, \mathbf{\Sigma}_{\mathbf{Y}}/R)$ for large R. Replacing $\mathbf{\Sigma}$ by $\mathbf{\Sigma}_{\mathbf{Y}}/R$ in the formulae from the previous frame, the confidence ellipsoid for the unknown mean $\mathbf{I}$ is centered at $\hat{\mathbf{Y}}_R$ with principal axis half-lengths $\sqrt{q_{1-\alpha}(\chi^2_d)\,\lambda_i/R}$, where λ_i are the eigenvalues of $\mathbf{\Sigma}_{\mathbf{Y}}$ (Eq. 1.24).
Since $\mathbf{\Sigma}_{\mathbf{Y}}$ is unknown, we estimate it by the **sample covariance matrix**:

Sample Covariance Matrix (Eq. 1.25)

$$\hat{\mathbf{S}}^2_R \;=\; \frac{1}{R-1}\sum_{k=1}^{R}\big(\mathbf{Y}_k - \hat{\mathbf{Y}}_R\big)\big(\mathbf{Y}_k - \hat{\mathbf{Y}}_R\big)^T.$$

This is the unbiased estimator of the covariance matrix $\mathbf{\Sigma}_{\mathbf{Y}}$ (analogous to the scalar sample variance $\hat{S}^2_R$ in the one-dimensional case). Its (i,j)-entry is the sample covariance between coordinates i and j: $\hat{\mathbf{S}}^2_R(i,j) = \frac{1}{R-1}\sum_{k=1}^{R}(Y_k^{(i)} - \hat{Y}_R^{(i)})(Y_k^{(j)} - \hat{Y}_R^{(j)})$. Here $Y_k^{(i)}$ (Y sub k superscript i) is the i-th component of the k-th replication.

4.1.2.3 Confidence Regions for the Estimator $\hat{\mathbf{Y}}_R$ II

Hotelling's T^2 (Remark 4.1.15). When the true covariance $\mathbf{\Sigma}_{\mathbf{Y}}$ is replaced by its estimate $\hat{\mathbf{S}}_R^2$, the statistic $R(\hat{\mathbf{Y}}_R - \mathbf{I})^T(\hat{\mathbf{S}}_R^2)^{-1}(\hat{\mathbf{Y}}_R - \mathbf{I})$ follows Hotelling's T^2-distribution (a multivariate generalisation of the Student t-distribution). For large R this converges to the χ_d^2 distribution, justifying the use of chi-square quantiles in constructing confidence regions when R is large.

4.1.2.3 Confidence Regions for the Estimator $\hat{\mathbf{Y}}_R$ III

Example 4.1.16 — Heights and Weights ($R = 1000$, $d = 2$)

We use the SOCR HeightsWeights dataset (first 1000 entries) to illustrate the multivariate confidence ellipsoid. Here $Y^{(1)}$ = weight (in lbs) and $Y^{(2)}$ = height (in inches).

Individual 95% confidence intervals for each component:

For weight: sample mean $\hat{Y}_R^{(1)} = 127.48$ lbs, sample standard deviation $\hat{S}_R^{(1)} = 11.80$ lbs. Standard error of mean: $11.80/\sqrt{1000} \approx 0.373$. CI:
$(127.48 - 1.96 \times 0.373,\ 127.48 + 1.96 \times 0.373) = (126.75,\ 128.21)$ lbs.

For height: sample mean $\hat{Y}_R^{(2)} = 68.05$ inches, sample standard deviation $\hat{S}_R^{(2)} = 1.94$ inches. Standard error: $1.94/\sqrt{1000} \approx 0.061$. CI: $(67.93,\ 68.17)$ inches.

Joint confidence ellipsoid:

$$\hat{\mathbf{S}}_R^2 = \begin{pmatrix} 139.29 & 11.40 \\ 11.40 & 3.75 \end{pmatrix}, \quad \lambda_1 = 140.24,\ \lambda_2 = 2.79.$$

The dominant eigenvector $\mathbf{f}_1 \approx (0.9965, 0.0832)^T$ points almost entirely in the weight direction — weight varies much more than height between individuals. With $q_{0.95}(\chi_2^2) = 5.991$ and $R = 1000$, the ellipsoid axis half-lengths are $\sqrt{5.991 \times 140.24/1000} \approx 0.916$ lbs (weight direction) and $\sqrt{5.991 \times 2.79/1000} \approx 0.129$ inches (height direction). The confidence ellipse is highly elongated along

4.1.2 Python: Multivariate Confidence Ellipsoid

```
import numpy as np
from scipy.stats import chi2

def conf_ellipsoid(Y_samples, alpha=0.05):
    """Confidence ellipsoid for the mean of d-dimensional i.i.d. samples."""
    R, d = Y_samples.shape
    Y_bar = Y_samples.mean(axis=0)              # sample mean (d-vector)

    diff = Y_samples - Y_bar
    S2   = (diff.T @ diff) / (R - 1)            # sample covariance matrix (d x d)
    eigvals, eigvecs = np.linalg.eigh(S2)       # eigendecomposition
    q = chi2.ppf(1 - alpha, df=d)               # chi-square quantile at level 1-alpha

    # Half-lengths of principal axes of the confidence ellipsoid
    # Each axis: sqrt(q * lambda_i / R) in direction eigvec[:,i]
    axis_lengths = np.sqrt(q * eigvals / R)
    return Y_bar, axis_lengths, eigvecs, S2

# Simulate 2D example: I = (2.0, 3.0), Sigma = [[1, 0.5],[0.5, 2]]
rng   = np.random.default_rng(42)
Sigma = np.array([[1., 0.5], [0.5, 2.]])
L     = np.linalg.cholesky(Sigma)               # Cholesky factor of Sigma
R     = 1000
Y     = (L @ rng.standard_normal((2, R))).T + np.array([2.0, 3.0])

Y_bar, axes, vecs, S2 = conf_ellipsoid(Y, alpha=0.05)
print(f"Estimated mean        : {Y_bar}")
```

4.2 Estimation of $k(I)$: Motivation and Setup I

In many practical applications, the quantity we care about is not the raw mean $I = \mathbb{E}Y$ but a nonlinear transformation of it. Section 4.2 asks: if we can estimate I well, how do we estimate $z = k(I)$ for some prescribed smooth function k, and what is the confidence interval?
Concrete examples where this arises:

- **Life insurance net premium:** The premium for a whole-life policy at force of interest r (r, a continuous interest rate) is $\Pi = r\,\mathbb{E}e^{-rT}/(1 - \mathbb{E}e^{-rT}) = k(I)$, where $I = \mathbb{E}e^{-rT}$ and T is the insured's future lifetime. We can simulate T values and average e^{-rT} to get $\hat{I}$, then plug into k to get $\hat{\Pi}$.
- **Standard deviation:** $k(I) = \sqrt{I}$ where $I = \mathbb{E}Y^2 - (\mathbb{E}Y)^2$ is the variance, estimated from simulation.
- **Ratio of means:** $z = I^{(1)}/I^{(2)}$, e.g. expected total cost divided by expected throughput, where each expectation is estimated separately.

4.2 Estimation of $k(I)$: Motivation and Setup II

Problem Setup and Natural Estimator

Let $k : \mathbb{R}^d \to \mathbb{R}$ be a prescribed continuous function (the transformation we apply), $\mathbf{I} = \mathbb{E}\mathbf{Y}$ the true d-dimensional mean, and $\hat{\mathbf{X}}_R$ a strongly consistent estimator of $\mathbf{I}$ (meaning $\hat{\mathbf{X}}_R \to \mathbf{I}$ with probability 1). The natural estimator of $z = k(\mathbf{I})$ is the *plug-in estimator* $\hat{Z}_R = k(\hat{\mathbf{X}}_R)$. The **asymptotic variance** of this estimator is defined as:

$$\varsigma^2 = \lim_{R\to\infty} R\,\mathrm{Var}(\hat{Z}_R).$$

Here ς (varsigma, a variant of sigma) is the asymptotic standard deviation constant, and $R\,\mathrm{Var}(\hat{Z}_R)$ is the variance scaled by R — which converges to a finite limit because variance decreases at rate $1/R$.

Important caveat: the plug-in estimator is generally biased. Even when $\hat{X}_R$ is an unbiased estimator of I, the transformation $k(\hat{X}_R)$ is generally *biased* for $k(I)$ because of the nonlinearity of k. For example, with $k(y) = y^2$: $\mathbb{E}[\hat{Y}_R^2] = (\mathbb{E}Y)^2 + \mathrm{Var}(\hat{Y}_R) = I^2 + \sigma_Y^2/R \neq I^2 = k(I)$. The estimator systematically overestimates I^2. However, this bias is of order $O(R^{-1})$, while the standard deviation is $O(R^{-1/2})$, so for large R the bias is negligible compared to the uncertainty. We can usually ignore the bias in practice.

4.2.1 The Delta Method: 1D Case I

The **delta method** is the key mathematical tool for deriving the confidence interval for $k(I)$. The core idea is: if k is smooth and $\hat{Y}_R \approx I$ for large R, we can approximate $k(\hat{Y}_R)$ by a linear (first-order Taylor) expansion around I, because close to I the nonlinear function k looks like a straight line.

4.2.1 The Delta Method: 1D Case II

Taylor Expansion Argument

If k is continuously differentiable near I (meaning the derivative k' exists and is continuous), then for x near I:

$$k(x) = k(I) + k'(I)(x - I) + o(x - I) \quad \text{as } x \to I.$$

Here $o(x - I)$ (little-o notation) means a remainder term that is smaller than $|x - I|$ for small $|x - I|$ (it vanishes faster than linearly). Substituting $x = \hat{Y}_R$ and multiplying both sides by $\sqrt{R}$:

$$\sqrt{R}\big(k(\hat{Y}_R) - k(I)\big) = k'(I) \cdot \underbrace{\sqrt{R}\,(\hat{Y}_R - I)}_{\xrightarrow{\mathcal{D}} \mathcal{N}(0,\sigma_Y^2)} + \underbrace{\sqrt{R} \cdot o(\hat{Y}_R - I)}_{\to 0 \text{ in probability}}.$$

The remainder vanishes: $\sqrt{R}\,|\hat{Y}_R - I|$ is bounded in probability (by the CLT), while the ratio $o(|\hat{Y}_R - I|)/|\hat{Y}_R - I| \to 0$. So the remainder $\sqrt{R} \cdot o(\hat{Y}_R - I) \to 0$ in probability.

4.2.1 The Delta Method: 1D Case III

Proposition 4.2.1 — Delta Method, 1D Version (Eq. 2.2)

Let $Y_1, Y_2, \ldots$ be i.i.d. with mean $I = \mathbb{E}Y$, variance $\sigma_Y^2 = \mathrm{Var}(Y) < \infty$, and let k be continuously differentiable at I with $k'(I) \neq 0$. Then:

$$\sqrt{R}\left(k(\hat{Y}_R) - k(I)\right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2), \quad \varsigma^2 = k'(I)^2\,\sigma_Y^2.$$

Here $k'(I)$ (k prime of I, the derivative of k evaluated at the true value I) acts as a magnification factor: where k is steep (large $|k'(I)|$), a small error in estimating I becomes a large error in $k(I)$; where k is flat (small $|k'(I)|$), errors in I barely affect $k(I)$. The factor σ_Y^2 is the variance of a single simulation output, measuring the intrinsic variability of the problem.

4.2.1 The Delta Method: 1D Case IV

Since both I and σ_Y^2 are unknown, we replace them by the consistent estimators $\hat{Y}_R$ and $\hat{S}_R^2$. By Slutsky's theorem (continuous functions of consistent estimators are consistent), this substitution preserves the asymptotic normal distribution, giving us a practical, computable confidence interval:

Practical CI for $k(I)$ at Level $1 - \alpha$ (Eq. 2.3)

$$\left(k(\hat{Y}_R) - z_{1-\alpha/2} \, \frac{\hat{S}_R \, |k'(\hat{Y}_R)|}{\sqrt{R}}, \;\; k(\hat{Y}_R) + z_{1-\alpha/2} \, \frac{\hat{S}_R \, |k'(\hat{Y}_R)|}{\sqrt{R}} \right). \tag{2.3}$$

Here $k(\hat{Y}_R)$ (k of Y-hat sub R) is the point estimate of $k(I)$; $z_{1-\alpha/2}$ is the normal quantile (1.96 for 95%); $\hat{S}_R$ is the sample standard deviation of the Y_j values; $|k'(\hat{Y}_R)|$ (absolute value of k prime of Y-hat sub R) is the estimated derivative of k at the estimated mean; and $\sqrt{R}$ is the square root of the number of replications.
The half-width $z_{1-\alpha/2} \, \hat{S}_R \, |k'(\hat{Y}_R)|/\sqrt{R}$ is wide when k is steep (large $|k'|$) and narrow when k is flat (small $|k'|$).

4.2.1 The Delta Method: 1D Case V

How to Read the Delta Method CI

The confidence interval (2.3) has the same form as the basic CI (1.9) for a mean, but with the standard deviation $\hat{S}_R$ replaced by $\hat{S}_R\,|k'(\hat{Y}_R)|$. The derivative factor $|k'(\hat{Y}_R)|$ propagates the uncertainty in $\hat{Y}_R$ through the nonlinear transformation k. A steep k amplifies errors; a flat k suppresses them. When $k'(I) = 0$ (the transformation is flat at the true value), the delta method gives a degenerate CI of zero width — in that case higher-order terms matter and the method does not apply directly.

4.2.1 The Delta Method: 1D Case VI

Example 4.2.2 — Life Insurance Net Premium

The net premium for whole-life assurance at continuous interest rate r (r, the force of interest, e.g. $r = 0.05$ for 5% per year) is:

$$\Pi = \frac{r\,\mathbb{E}e^{-rT}}{1 - \mathbb{E}e^{-rT}} = k(I),$$

where $I = \mathbb{E}e^{-rT}$ and T is the future lifetime of the insured (a random variable). The factor e^{-rT} (e to the power minus r times T) is the discount factor — how much a payment of 1 at time T is worth today at interest rate r.

We set $Y = e^{-rT}$, simulate many lifetimes $T_1, \ldots, T_R$, and average e^{-rT_j} to get $\hat{Y}_R \approx I$. The transformation is $k(x) = rx/(1-x)$, with derivative: $k'(x) = r/(1-x)^2$ (applying the quotient rule). The delta method CI (2.3) for the premium $\Pi = k(I)$ is:

$$k(\hat{Y}_R) \;\pm\; \frac{z_{1-\alpha/2}\, r\, \hat{S}_R}{(1-\hat{Y}_R)^2\,\sqrt{R}}.$$

Notice the $(1-\hat{Y}_R)^2$ term in the denominator. As $\hat{Y}_R \to 1$ (which happens when r is large or the expected lifetime T is long, so e^{-rT} is close to 1), the factor $(1-\hat{Y}_R)^2$ becomes very small, causing the CI to widen dramatically. This reflects the mathematical fact that the premium $\Pi = rx/(1-x)$ is a

4.2.1 The Multidimensional Delta Method I

When $\mathbf{I} = \mathbb{E}\mathbf{Y}$ is a d-dimensional vector and $k : \mathbb{R}^d \to \mathbb{R}$ maps it to a scalar, the delta method generalizes using the *gradient* ∇k (nabla k, also written grad k): the vector of all partial derivatives of k.

4.2.1 The Multidimensional Delta Method II

Two-Dimensional Case (Proposition 4.2.3, Eq. 2.5–2.6)

Let $\mathbf{Y}_j = (Y_j^{(1)}, Y_j^{(2)})^T$ (a 2-component random vector), $\mathbf{I} = (I^{(1)}, I^{(2)})^T$, and $k(y_1, y_2)$ continuously differentiable at $\mathbf{I}$. The Taylor linearization becomes:

$$k(\hat{\mathbf{Y}}_R) - k(\mathbf{I}) \approx k'_{y_1}(\hat{Y}_R^{(1)} - I^{(1)}) + k'_{y_2}(\hat{Y}_R^{(2)} - I^{(2)}),$$

where $k'_{y_i} = \partial k / \partial y_i \big|_{\mathbf{y}=\mathbf{I}}$ (the partial derivative of k with respect to y_i, evaluated at the true value $\mathbf{I}$). By the multivariate CLT and the delta method:

$$\sqrt{R}\,(k(\hat{\mathbf{Y}}_R) - k(\mathbf{I})) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2),$$

where the asymptotic variance ς^2 is (Eq. 2.6):

$$\varsigma^2 = (k'_{y_1})^2 \mathbf{\Sigma}_{\mathbf{Y}}(1,1) + (k'_{y_2})^2 \mathbf{\Sigma}_{\mathbf{Y}}(2,2) + 2k'_{y_1}k'_{y_2}\mathbf{\Sigma}_{\mathbf{Y}}(1,2). \tag{2.6}$$

Here $\mathbf{\Sigma}_{\mathbf{Y}}(i,j) = \mathrm{Cov}(Y^{(i)}, Y^{(j)})$ is the (i,j)-entry of the covariance matrix. The first two terms are the individual variance contributions (squared derivative times individual variance), and the third term is the cross-term arising from the correlation between components.

4.2.2 General Consistent Estimator and Asymptotic Bias I

The delta method extends beyond the sample mean to any strongly consistent estimator $\hat{X}_R \to I$ satisfying two conditions:

1. $\sqrt{R}(\hat{X}_R - \mathbb{E}\hat{X}_R) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2)$ (asymptotically normal when centered at its *own* mean, even if that mean is not exactly I).
2. $\sqrt{R}(\mathbb{E}\hat{X}_R - I) \to 0$ as $R \to \infty$ (the bias $\mathbb{E}\hat{X}_R - I$ vanishes faster than $1/\sqrt{R}$).

Under (A1) and (A2) together, $\sqrt{R}(\hat{X}_R - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2)$, and the delta method gives $\sqrt{R}(k(\hat{X}_R) - k(I)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, k'(I)^2\sigma^2)$.

Asymptotic Bias of $\hat{Z}_R = k(\hat{X}_R)$ (Eq. 2.17)

Under the stronger assumption (A2′): $R(\mathbb{E}\hat{X}_R - I) \to \beta^X$ (the bias of $\hat{X}_R$ is of exact order $1/R$ with constant β^X), the order-$1/R$ asymptotic bias of $\hat{Z}_R = k(\hat{X}_R)$ is:

$$\beta^Z = \beta^X\, k'(I) + \frac{\sigma^2\, k''(I)}{2}. \tag{2.17}$$

Here $k''(I)$ (k double-prime of I) is the second derivative of k evaluated at the true value I. The first term $\beta^X k'(I)$ simply propagates any existing bias of $\hat{X}_R$ through the transformation k. The second term $\sigma^2 k''(I)/2$ is the *Jensen's inequality effect*: even if $\hat{X}_R$ is perfectly unbiased ($\beta^X = 0$), the curvature of k (measured by k'') induces a bias in $k(\hat{X}_R)$.

A bias-corrected estimator is $k(\hat{X}_R) - \hat{\beta}^Z/R$, but this correction is usually unnecessary because the bias $O(R^{-1})$ is dominated by the standard deviation $O(R^{-1/2})$ for large R.

4.2.3 Mean Square Error, Bias, and Variance (Eq. 2.19)

The *mean square error* (MSE) of any estimator $\hat{X}_R$ of I decomposes into two parts:

$$\mathrm{MSE}(\hat{X}_R) = \mathbb{E}(\hat{X}_R - I)^2 = \underbrace{\mathrm{Var}(\hat{X}_R)}_{\text{random fluctuation}} + \underbrace{(\mathbb{E}\hat{X}_R - I)^2}_{\text{squared systematic bias}} .$$

Variance captures how much the estimator fluctuates randomly around its own mean. Squared bias captures how far the estimator's mean is from the truth. An unbiased estimator has MSE = Variance. In general, there is a bias-variance tradeoff: techniques that reduce bias (like using $\hat{S}_R^2$ instead of $\hat{\sigma}_R^2$) may increase variance.

For large R, using $\lim_{R\to\infty} R\,\mathrm{Var}(\hat{X}_R) = \varsigma^2$ and $\lim_{R\to\infty} R(\mathbb{E}\hat{X}_R - I) = \beta$:

$$\mathrm{MSE}(\hat{X}_R) \approx \frac{\varsigma^2}{R} + \frac{\beta^2}{R^2}.$$

4.2.2 General Consistent Estimator and Asymptotic Bias IV

The variance term is $O(R^{-1})$ and the squared bias is $O(R^{-2})$ — the bias term is negligible for large R, justifying the common practice of ignoring bias corrections.

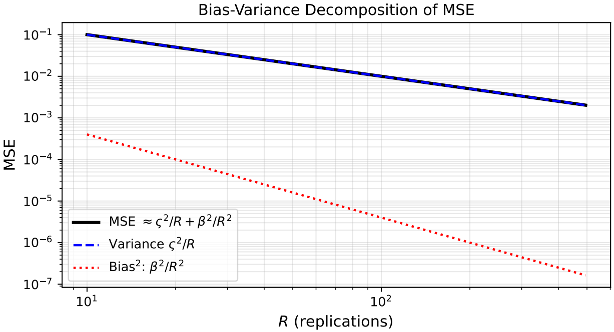
Example 4.2.9 — $k(y) = y^2$, bias verification

Let $\hat{Z}_R = \hat{Y}_R^2$. The derivatives are $k'(y) = 2y$ and $k''(y) = 2$. Since the sample mean $\hat{Y}_R$ is unbiased ($\beta^X = 0$), formula (2.17) gives:

$$\beta^Z = 0 + \frac{\sigma_Y^2 \cdot 2}{2} = \sigma_Y^2.$$

We can verify this directly: $\mathbb{E}(\hat{Y}_R)^2 = (\mathbb{E}\hat{Y}_R)^2 + \mathrm{Var}(\hat{Y}_R) = I^2 + \sigma_Y^2/R = k(I) + \sigma_Y^2/R$. So $\hat{Y}_R^2$ overestimates I^2 by σ_Y^2/R on average — this is the Jensen's inequality effect: $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$. The delta method asymptotic variance: $\varsigma^2 = (2I)^2\sigma_Y^2 = 4I^2\sigma_Y^2$.
For 95% CI: $\hat{Y}_R^2 \pm 1.96 \times 2|\hat{Y}_R|\,\hat{S}_R/\sqrt{R}$.

4.2.2 General Consistent Estimator and Asymptotic Bias V



4.2 Python: Delta Method Confidence Interval

```python
import numpy as np
from scipy import stats

def delta_method_ci(Y_samples, k_func, k_prime, alpha=0.05):
    """
    Compute CI for k(I) using the delta method.
    Y_samples : 1D array of i.i.d. realizations of Y (E[Y]=I)
    k_func    : callable, the transformation k
    k_prime   : callable, derivative k' (evaluated at a scalar)
    """
    R      = len(Y_samples)
    Y_bar = Y_samples.mean()
    S      = Y_samples.std(ddof=1)    # sample std dev (ddof=1: divide by R-1)

    z_est     = k_func(Y_bar)                              # point estimate of k(I)
    asymp_std = abs(k_prime(Y_bar)) * S / np.sqrt(R)  # estimated half-width / z

    z_q = stats.norm.ppf(1 - alpha / 2)   # normal quantile; 1.96 for 95%
    return z_est, z_est - z_q * asymp_std, z_est + z_q * asymp_std

# Example: k(y) = y^2. Y ~ N(0.5, 0.3^2), so I=0.5 and k(I)=0.25.
# Delta method: varsigma^2 = (2*0.5)^2 * 0.09 = 0.09; std ~ 0.3/sqrt(R).
rng = np.random.default_rng(42)
Y   = rng.normal(0.5, 0.3, size=1000)
est, lo, hi = delta_method_ci(
    Y, k_func=lambda y: y**2, k_prime=lambda y: 2*y)
print(f"k(I) estimate: {est:.4f}")
```

4.3 Estimating a Quantile: Motivation and Setup I

So far in this chapter we have estimated *means*: $I = \mathbb{E}X$, the theoretical average of a random variable. A **quantile** is a fundamentally different summary of a distribution. The p-th quantile (for p (p) in the open interval $(0, 1)$) is the value below which exactly a fraction p of the probability mass lies.

Quantiles arise everywhere in practice:

- In **financial risk management**, the 99th percentile ($p = 0.99$) of the distribution of daily portfolio losses is called the **Value-at-Risk (VaR)**. Banks are required by regulation to hold capital against this number: "our loss will exceed this threshold on at most 1% of trading days."
- In **reliability engineering**, the 10th percentile ($p = 0.10$) of a component's lifetime distribution is the time by which 10% of components will have failed — a pessimistic but conservative design target.
- In **healthcare**, the 95th percentile of a biomarker distribution defines the upper limit of a "normal" reference range.

4.3 Estimating a Quantile: Motivation and Setup II

Because quantiles depend on the *shape* of the entire distribution (not just its mean), estimating them requires different tools than those developed in Sections 4.1 and 4.2.

Definition: The p-th Quantile (Eq. 3.1)

Let X be a real-valued random variable with cumulative distribution function $F(x) = \mathbb{P}(X \leq x)$. The p**-th quantile** (also called the p-th percentile when multiplied by 100) is the generalized inverse:

$$q_p \;=\; F^{\leftarrow}(p) \;=\; \inf\{x \in \mathbb{R} : F(x) \geq p\}.$$

The symbol $\inf$ means "infimum" (greatest lower bound). For a **continuous** distribution function F that is strictly increasing, $F^{\leftarrow}(p) = F^{-1}(p)$ is simply the unique solution to the equation $F(q_p) = p$. In other words, q_p is the value such that $\mathbb{P}(X \leq q_p) = p$ exactly.

4.3 Estimating a Quantile: Motivation and Setup III

Notation Explained

F (capital F) — the **cumulative distribution function** (CDF) of X; $F(x)$ gives the probability that X does not exceed x. p (p) — the **quantile level**, a number strictly between 0 and 1; for example $p = 0.95$ means the 95th percentile. q_p (q sub p) — the p**-th quantile**: the threshold exceeded with probability $1 - p$. $f(x)$ (lowercase f) — the **probability density function** (PDF) of X; if it exists, $f(x) = F'(x)$ (the derivative of F).

4.3 Estimating a Quantile: Motivation and Setup IV

Given an i.i.d. (independent and identically distributed) sample $X_1, X_2, \ldots, X_R$ drawn from distribution F, we estimate q_p using the **sample quantile** (also called the *empirical quantile*).
The first step is to **sort** the observations from smallest to largest:

$$X_{(1)} \ \leq \ X_{(2)} \ \leq \ \cdots \ \leq \ X_{(R)}.$$

These sorted values are called **order statistics**. Here $X_{(k)}$ (X sub (k) in parentheses) denotes the k-th smallest observation. For instance, $X_{(1)}$ is the minimum and $X_{(R)}$ is the maximum.

4.3 Estimating a Quantile: Motivation and Setup V

Sample Quantile Estimator

The empirical quantile function $\hat{F}_R^{\leftarrow}(p)$ is defined as:

$$\hat{F}_R^{\leftarrow}(p) = \begin{cases} X_{(\lfloor pR \rfloor)} & \text{if } pR \text{ is a whole number,} \\ X_{(\lfloor pR \rfloor + 1)} & \text{otherwise.} \end{cases}$$

Here $\lfloor pR \rfloor$ (the **floor** of pR, read "floor of p-times-R") means the largest integer that does not exceed pR. For example, if $R = 100$ and $p = 0.9$, then $pR = 90$ is a whole number, so the estimator is $X_{(90)}$ — the 90th smallest observation. If $R = 95$ and $p = 0.9$, then $pR = 85.5$ and $\lfloor 85.5 \rfloor = 85$, so the estimator is $X_{(86)}$ — the 86th smallest observation.

For the theoretical analysis that follows, we use the simplified estimator $\hat{q}_p = X_{(\lfloor pR \rfloor)}$, which has the same asymptotic properties.

4.3 Estimating a Quantile: Motivation and Setup VI

Intuition: Why the Order Statistic?

In a sample of R observations, approximately pR of them should fall below the true quantile q_p (since each observation independently has probability p of lying below q_p). So the (pR)-th smallest observation is a natural estimate of the value that separates the bottom fraction p from the top fraction $1 - p$ of the distribution.

4.3.1 Asymptotic Normality of the Sample Quantile I

One of the most important results in statistics is that the sample quantile $\hat{q}_p = X_{(\lfloor pR \rfloor)}$ is not only consistent (it converges to the true q_p as $R \to \infty$) but also **asymptotically normal**. This means that for large R, $\hat{q}_p$ has approximately a normal (bell-curve) distribution centred at the true q_p. This normality allows us to construct confidence intervals using the same normal-quantile machinery we used for the sample mean in Section 4.1.

4.3.1 Asymptotic Normality of the Sample Quantile II

Proposition 4.3.1 — Asymptotic Normality of the Sample Quantile (Eq. 3.2)

Assume the distribution F has a density f (lowercase f) that is continuous and strictly positive in an open neighbourhood of q_p (i.e., $f(q_p) > 0$). Then:

1. **Strong consistency:** $\hat{q}_p \to q_p$ with probability 1 as $R \to \infty$. Running more simulations eventually pins down the quantile exactly.
2. **Asymptotic normality:** as $R \to \infty$,

$$\sqrt{R}\,(\hat{q}_p - q_p) \xrightarrow{\mathcal{D}} \mathcal{N}(0,\, \varsigma^2)\,, \qquad \text{where } \varsigma^2 = \frac{p(1-p)}{f(q_p)^2}.$$

The symbol $\xrightarrow{\mathcal{D}}$ means "converges in distribution to." $\mathcal{N}(0, \varsigma^2)$ (script N) is the normal distribution with mean 0 and variance ς^2 (varsigma squared).

4.3.1 Asymptotic Normality of the Sample Quantile III

Notation for the Asymptotic Variance $\varsigma^2 = p(1-p)/f(q_p)^2$

ς^2 (varsigma squared, sometimes written σ_q^2) — the **asymptotic variance constant**; when we write $\sqrt{R}(\hat{q}_p - q_p) \to \mathcal{N}(0, \varsigma^2)$, the actual variance of $\hat{q}_p$ is approximately ς^2/R for large R.

p (p) — the quantile level in $(0, 1)$.

$1 - p$ (one minus p) — the complementary probability; together, $p(1-p)$ behaves like the variance of a Bernoulli(p) random variable and is maximised at $p = 0.5$ (the median).

$f(q_p)$ (f of q-sub-p) — the **density of X evaluated at the quantile** q_p. This is the height of the probability density curve at the point we are trying to estimate.

4.3.1 Asymptotic Normality of the Sample Quantile IV

How to Read the Formula $\varsigma^2 = p(1-p)/f(q_p)^2$

Think of $f(q_p)$ as measuring how "crowded" the distribution is near q_p. When $f(q_p)$ is **large** (the density has a sharp peak near q_p, so many observations fall close to q_p), the denominator $f(q_p)^2$ is large and ς^2 is small — the sample quantile is easy to estimate accurately.

When $f(q_p)$ is **small** (the distribution is spread out near q_p, with few observations falling close to it), ς^2 is large and estimation is imprecise. This happens especially in the tails of the distribution: at extreme quantiles like $p = 0.999$, the density $f(q_{0.999})$ is very small, making the tail quantile hard to pin down.

The numerator $p(1-p)$ is the binomial variance from asking "did X_i fall below q_p?" (yes with probability p, no with probability $1-p$). It peaks at the median ($p = 0.5$, where $p(1-p) = 0.25$) and shrinks toward the tails, but the denominator $f(q_p)^2$ typically shrinks even faster, making tail quantiles harder overall.

4.3.1 Asymptotic Normality of the Sample Quantile V

Proof sketch of Proposition 4.3.1. The argument uses two steps. Understanding these steps gives genuine insight into *why* the formula is what it is.

Step 1 — Uniform order statistics (Lemma 4.3.2). If $U_1, U_2, \ldots, U_R$ are i.i.d. $\mathcal{U}[0,1]$ (uniform on the interval $[0,1]$), the k-th order statistic $U_{(k)}$ (the k-th smallest uniform value) follows a Beta$(k, R-k+1)$ distribution (Beta distribution with parameters k and $R-k+1$). The Beta distribution has mean $k/(R+1)$ and variance $k(R-k+1)/[(R+1)^2(R+2)]$.
Setting $k = \lfloor Rp \rfloor$ and applying the CLT to this Beta distribution (using the fact that $\mathrm{Beta}(k, R-k+1)$ is the ratio of two Erlang sums, which are themselves sums of exponential random variables):

$$\sqrt{R}\,(U_{(\lfloor Rp \rfloor)} - p) \xrightarrow{\mathcal{D}} \mathcal{N}(0,\ p(1-p)).$$

This says: the fraction of "successes" among order statistics is approximately normal with variance $p(1-p)/R$ — exactly the binomial variance we expect from asking "is $U_i \leq p$?"

4.3.1 Asymptotic Normality of the Sample Quantile VI

Step 2 — Transfer to a general distribution via the quantile transform. Any random variable X with CDF F can be written as $X \overset{d}{=} F^{-1}(U)$ where $U \sim \mathcal{U}[0,1]$ (the **quantile transform**). Therefore $X_{(\lfloor Rp \rfloor)} \overset{d}{=} F^{-1}(U_{(\lfloor Rp \rfloor)})$. Expanding F^{-1} around p using a Taylor (first-order) expansion:

$$F^{-1}(p+h) \;\approx\; F^{-1}(p) + h \cdot \frac{d}{dp}F^{-1}(p) = q_p + h \cdot \frac{1}{f(q_p)},$$

where the last equality uses the fact that the derivative of the quantile function $F^{-1}(p)$ equals $1/f(F^{-1}(p)) = 1/f(q_p)$ (this follows from differentiating $F(F^{-1}(p)) = p$ with respect to p, giving $f(F^{-1}(p)) \cdot (F^{-1})'(p) = 1$).
Setting $h = U_{(\lfloor Rp \rfloor)} - p$ and multiplying by $\sqrt{R}$:

$$\sqrt{R}(\hat{q}_p - q_p) \;\approx\; \frac{1}{f(q_p)} \cdot \underbrace{\sqrt{R}(U_{(\lfloor Rp \rfloor)} - p)}_{\to \mathcal{N}(0,\, p(1-p))} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0,\ \frac{p(1-p)}{f(q_p)^2}\right).$$

The $1/f(q_p)$ factor stretches the variance by $(1/f(q_p))^2$, yielding $\varsigma^2 = p(1-p)/f(q_p)^2$. □

Worked Numerical Example: $X \sim \mathrm{Exp}(1)$, $p = 0.9$

The exponential distribution with rate 1 (mean 1) has:

$$F(x) = 1 - e^{-x} \quad \text{and} \quad f(x) = e^{-x}, \quad x \geq 0.$$

Step 1 — Find the true quantile $q_{0.9}$. Solve $F(q_{0.9}) = 0.9$: $1 - e^{-q_{0.9}} = 0.9$, so $e^{-q_{0.9}} = 0.1$, giving $q_{0.9} = -\ln(0.1) = \ln(10) \approx 2.3026$.
Step 2 — Evaluate the density at the quantile. $f(q_{0.9}) = e^{-q_{0.9}} = e^{-\ln(10)} = 1/10 = 0.1$.
Step 3 — Compute the asymptotic variance.

$$\varsigma^2 = \frac{p(1-p)}{f(q_p)^2} = \frac{0.9 \times 0.1}{(0.1)^2} = \frac{0.09}{0.01} = 9.$$

Step 4 — Standard error and CI for $R = 1{,}000$ replications. The standard error of $\hat{q}_{0.9}$ is approximately: $\varsigma/\sqrt{R} = 3/\sqrt{1000} \approx 0.0949$.
For a 95% confidence interval ($z_{0.975} = 1.96$): half-width $= 1.96 \times 0.0949 \approx 0.186$. CI: approximately $(2.3026 - 0.186,\ 2.3026 + 0.186) = (2.117,\ 2.489)$.
Interpretation. The asymptotic variance $\varsigma^2 = 9$ is large because the density at the 90th percentile ($f = 0.1$) is small — the exponential distribution is quite spread out at its 90th percentile. We need

4.3.1 Building the Confidence Interval: Estimating the Density $f(q_p)$ I

The asymptotic normality result (Proposition 4.3.1) gives us a confidence interval for q_p, but only if we know the asymptotic standard deviation $\varsigma = \sqrt{p(1-p)}/f(q_p)$. The numerator $\sqrt{p(1-p)}$ is known exactly because p is specified by us. The denominator $f(q_p)$ is unknown and must be estimated from the data.

The key mathematical insight is that the **quantile function** $Q(p) = F^{-1}(p)$ satisfies a beautiful relationship between its derivative and the density. Differentiating $F(F^{-1}(p)) = p$ with respect to p gives:

$$f(q_p) \cdot Q'(p) = 1 \quad \Longrightarrow \quad Q'(p) = \frac{1}{f(Q(p))} = \frac{1}{f(q_p)}.$$

So the quantity we need, $1/f(q_p)$, is simply the **derivative of the quantile function** $Q'(p)$ at the point p.

This turns the problem of density estimation into a problem of **derivative estimation**, which can be done via a finite-difference approximation:

$$Q'(p) \approx \frac{Q(p + m/R) - Q(p - m/R)}{2m/R},$$

4.3.1 Building the Confidence Interval: Estimating the Density $f(q_p)$ II

for a small "bandwidth" m (m, a positive integer). We replace the true quantile values $Q(p \pm m/R)$ with their empirical counterparts — the corresponding order statistics.

Order-Statistic Estimator of $Q'(p) = 1/f(q_p)$ (Proposition 4.3.4, Eq. 3.3)

Given sorted observations $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(R)}$, choose a bandwidth integer $m = m_R$ (m sub R, depending on R). Define:

$$\hat{Q}'_{p,R} \;=\; \frac{R}{2m}\Big(X_{(\lfloor pR \rfloor + m)} - X_{(\lfloor pR \rfloor - m + 1)}\Big). \tag{3.3}$$

How to read this: The upper order statistic $X_{(\lfloor pR \rfloor + m)}$ estimates the quantile $Q(p + m/R)$ (the level slightly above p). The lower order statistic $X_{(\lfloor pR \rfloor - m + 1)}$ estimates $Q(p - m/R)$ (the level slightly below p). Dividing their difference by $2m/R$ approximates the derivative $Q'(p)$. The prefactor $R/(2m)$ equals $1/(2m/R)$, the reciprocal of the step size.

4.3.1 Building the Confidence Interval: Estimating the Density $f(q_p)$ III

Notation

m_R (m sub R) — the **bandwidth**, a positive integer that controls the width of the finite-difference window; it must grow as R grows, but more slowly than R itself.

$X_{(\lfloor pR \rfloor + m)}$ — the order statistic at index $\lfloor pR \rfloor + m$; this is m positions to the *right* (larger side) of the sample quantile.

$X_{(\lfloor pR \rfloor - m + 1)}$ — the order statistic at index $\lfloor pR \rfloor - m + 1$; this is approximately m positions to the *left* (smaller side) of the sample quantile.

Consistency of the Estimator (Proposition 4.3.4)

If f has a continuous derivative near q_p (the density curve is smooth at the quantile) and the bandwidth satisfies:

$$m_R \to \infty \quad \text{and} \quad m_R = o(R) \quad \text{as } R \to \infty,$$

(meaning: m_R grows without bound, but grows strictly slower than R), then $\hat{Q}'_{p,R} \to 1/f(q_p)$ with probability 1. In plain terms: as long as we use a window that grows but not too fast, the estimator converges to the true value of $1/f(q_p)$.

Choosing the bandwidth m_R — two options:

- **Simple default rule:** $m_R = \lfloor \sqrt{R} \rfloor$ (floor of the square root of R). This satisfies both conditions automatically: $\sqrt{R} \to \infty$ and $\sqrt{R}/R = 1/\sqrt{R} \to 0$. For $R = 1{,}000$: $m = 31$. For $R = 10{,}000$: $m = 100$. This is the recommended starting point.

4.3.1 Building the Confidence Interval: Estimating the Density $f(q_p)$ V

- **MSE-optimal rule (Bloch–Gastwirth):** $m_R \propto R^{4/5}$ (R to the power four-fifths). This minimises the asymptotic Mean Squared Error (MSE) of $\hat{Q}'_{p,R}$, giving MSE $\sim C \cdot R^{-2/5}$. The rate $R^{-2/5}$ is slower than $R^{-1/2}$ (the usual Monte Carlo rate), reflecting the fundamental difficulty of density estimation from samples. In practice, the constant C depends on the unknown second derivative of f near q_p, so this optimal bandwidth is mainly of theoretical interest.

Confidence Interval for q_p (combining Propositions 4.3.1 and 4.3.4)

Substituting $\hat{Q}'_{p,R}$ as our estimate of $1/f(q_p)$ into the asymptotic normality result:

$$\hat{q}_p \;\pm\; z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{R}}\, \hat{Q}'_{p,R},$$

where $z_{1-\alpha/2}$ (z sub one-minus-alpha-over-two) is the standard normal quantile (for 95% CI: $z_{0.975} = 1.96$), $p(1-p)$ (p times one-minus-p) is the binomial variance factor, and $\hat{Q}'_{p,R}$ estimates $1/f(q_p)$. This confidence interval has coverage converging to $1-\alpha$ as $R \to \infty$.

4.3.1 Building the Confidence Interval: Estimating the Density $f(q_p)$ VI

Worked Numerical Example: $X \sim \mathrm{Exp}(1)$, $p = 0.9$, $R = 1{,}000$

We have $R = 1000$ i.i.d. observations from $\mathrm{Exp}(1)$. Sort them to get $X_{(1)} \leq \cdots \leq X_{(1000)}$. Set $p = 0.9$ and use bandwidth $m = \lfloor \sqrt{1000} \rfloor = 31$.

Sample quantile index: $\lfloor 0.9 \times 1000 \rfloor = 900$, so $\hat{q}_{0.9} = X_{(900)}$.

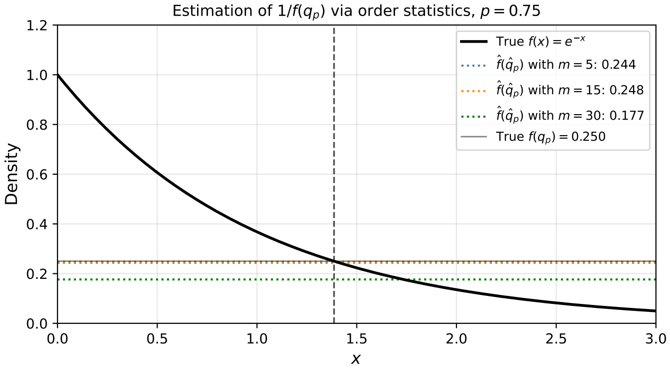
Estimate of $1/f(q_p)$: $\hat{Q}'_{0.9,1000} = \frac{1000}{2\times 31}\left(X_{(931)} - X_{(870)}\right) = \frac{1000}{62}\left(X_{(931)} - X_{(870)}\right)$.

In a typical simulation, one might find $X_{(870)} \approx 2.21$ and $X_{(931)} \approx 2.42$, giving $\hat{Q}'_{0.9,1000} \approx \frac{1000}{62} \times 0.21 \approx 3.39$. (The true value is $1/f(q_{0.9}) = 1/0.1 = 10$; with $m = 31$ the estimate is imprecise — this is the inherent difficulty of density estimation — but it improves as R grows.)

CI half-width: $1.96 \times \sqrt{0.9 \times 0.1/1000} \times 3.39 \approx 1.96 \times 0.0300 \times 3.39 \approx 0.199$.

Confidence interval: $(X_{(900)} - 0.199,\ X_{(900)} + 0.199)$. If $X_{(900)} \approx 2.30$, this gives approximately $(2.10,\ 2.50)$, which does contain the true value $\ln 10 \approx 2.303$.

4.3.1 Building the Confidence Interval: Estimating the Density $f(q_p)$ VII



4.3 Python: Sample Quantile and Confidence Interval

```
import numpy as np
from scipy import stats

def quantile_ci(X_samples, p, m=None, alpha=0.05):
    """
    Estimate the p-quantile and construct a (1-alpha) confidence interval.

    Parameters
    ----------
    X_samples : 1D numpy array of i.i.d. observations from unknown CDF F
    p         : float in (0, 1), the desired quantile level
    m         : int, bandwidth for density estimate (default: floor(sqrt(R)))
    alpha     : float, significance level (default: 0.05 for a 95% CI)

    Returns
    -------
    q_hat : float -- sample quantile estimate
    lo    : float -- lower confidence limit
    hi    : float -- upper confidence limit
    """
    R        = len(X_samples)
    X_sorted = np.sort(X_samples)         # sort ascending: X_(1) <= ... <= X_(R)
    idx      = int(np.floor(p * R))       # index of sample quantile (0-based)
    q_hat    = X_sorted[idx]              # sample quantile = X_(floor(pR))

    if m is None:
        m = max(1, int(np.floor(np.sqrt(R))))   # simple bandwidth: sqrt(R)
```

4.4.1 The Chernoff Bound: Exponential Concentration I

In Section 4.1 we compared two ways to control absolute error: Chebyshev's inequality and the CLT. Now we study a third, sharper tool called the **Chernoff bound**, which gives exponentially small error probabilities. Unlike Chebyshev (which gives bounds that shrink as $1/R$, a "polynomial" rate), the Chernoff bound gives bounds that shrink as e^{-cR} (an "exponential" rate in R). For even moderately large R, this exponential decay is astronomically smaller than any polynomial decay.
The Chernoff bound applies specifically when the random variables Y_i are Bernoulli (each Y_i is either 0 or 1), which is the case for many Monte Carlo estimators — for example, the hit-or-miss estimator for computing areas or probabilities.

Setup: Bernoulli Sums

Let $Y_1, Y_2, \ldots, Y_R$ be i.i.d. Bernoulli(p) random variables. This means each Y_i equals 1 with probability p (p, the success probability) and equals 0 with probability $1-p$. The sum $S_R = Y_1 + Y_2 + \cdots + Y_R$ counts the total number of "successes" in R trials. The expectation of S_R is $\mathbb{E}S_R = Rp$, and the sample mean $\hat{Y}_R = S_R/R$ estimates p.

4.4.1 The Chernoff Bound: Exponential Concentration II

Proposition 4.4.1 — The Chernoff Bound (Eqs. 4.1–4.3)

For any $\varepsilon \in (0,1)$ (epsilon, the relative error tolerance):

Upper tail (overestimation):

$$\mathbb{P}(S_R \geq (1+\varepsilon)\,Rp) \;\leq\; \exp\left(-\frac{\varepsilon^2 Rp}{2+\varepsilon}\right) \;\leq\; \exp\left(-\frac{\varepsilon^2 Rp}{3}\right). \tag{4.1}$$

Lower tail (underestimation):

$$\mathbb{P}(S_R \leq (1-\varepsilon)\,Rp) \;\leq\; \exp\left(-\frac{\varepsilon^2 Rp}{2}\right). \tag{4.2}$$

Two-sided (combining both tails):

$$\mathbb{P}(|S_R - Rp| \geq \varepsilon\,Rp) \;\leq\; 2\exp\left(-\frac{\varepsilon^2 Rp}{3}\right). \tag{4.3}$$

Here $\exp(x)$ (exponential function) means e^x where $e \approx 2.718$. The condition $|S_R - Rp| \geq \varepsilon Rp$ is equivalent to $|\hat{Y}_R - p| \geq \varepsilon p$ — a **relative error** of at least ε in the estimate of p.

4.4.2 Absolute vs Relative Error: A Striking Contrast I

We now compare the sample sizes required for absolute and relative error control, using the concrete case of i.i.d. Bernoulli(p) observations ($Y_i \in \{0,1\}$, $\mathbb{E}Y_i = p$, $\mathrm{Var}(Y_i) = p(1-p)$). We fix $\varepsilon = \delta = 0.01$ throughout (1% error tolerance, 99% confidence).

Absolute error first. We want $\mathbb{P}(|\hat{Y}_R - p| \geq \varepsilon) \leq \delta$, i.e., the absolute discrepancy between our estimate and the truth is at most 0.01 with probability at least 99%.
Using the CLT formula (Eq. 1.7) with $\sigma_Y^2 = \mathrm{Var}(Y) = p(1-p)$, $z_{0.995} = 2.576$ (the 99.5th percentile of the standard normal, since $1 - \delta/2 = 1 - 0.005 = 0.995$):

$$R^{\mathrm{abs}} = z_{1-\delta/2}^2 \cdot \frac{p(1-p)}{\varepsilon^2} = 2.576^2 \cdot \frac{p(1-p)}{(0.01)^2} \approx 6{,}634 \cdot p(1-p).$$

$p = I$ (true probability)	$p(1-p)$ (variance)	R^{abs}	Interpretation
0.5	0.250	$\approx 1{,}659$	common probability
0.1	0.090	≈ 597	moderately rare
0.01	0.0099	≈ 66	rare
0.001	≈ 0.001	≈ 7	very rare

4.4.2 Absolute vs Relative Error: A Striking Contrast II

This is counterintuitive: *smaller* p requires *fewer* replications for absolute error control! For $p = 0.001$, just 7 replications suffice (theoretically) to ensure the estimate is within ± 0.01 of the truth.

4.4.2 Absolute vs Relative Error: A Striking Contrast III

Why absolute error control fails for rare events. Consider the extreme case $p = 0.001$ and a single replication ($R = 1$). The estimator $\hat{Y}_1$ takes only two values: 0 (with probability 0.999) or 1 (with probability 0.001). The outcome $\hat{Y}_1 = 0$ (which happens 99.9% of the time) gives $|\hat{Y}_1 - p| = |0 - 0.001| = 0.001 \leq 0.01 = \varepsilon$. So the absolute error condition is satisfied! But the estimate (0) tells us essentially nothing about the true value (0.001) — we only know it is "small," not by how much.

Key Insight: Absolute Error Is Misleading for Rare Events

When p is small, an estimate of 0 ("event never occurred") is within $\varepsilon = 0.01$ of the truth in absolute terms. But it gives zero information about the *magnitude* of p. The absolute error criterion (E1) is satisfied trivially, yet the estimate is practically useless. This is why we need relative error control for rare events.

Relative error corrects this. We want $\mathbb{P}(|\hat{Y}_R/p - 1| \geq \varepsilon) \leq \delta$, i.e., the estimate must be within $\pm 1\%$ of the true value *in percentage terms*. An estimate of 0 when $p = 0.001$ has *100% relative error* and fails this condition catastrophically.

Using Corollary 4.4.2 (the Chernoff bound, Eq. 4.4) with $\varepsilon = \delta = 0.01$:

$$R \geq \frac{3\ln(2/0.01)}{(0.01)^2 \cdot p} = \frac{3\ln(200)}{0.0001 \cdot p} = \frac{3 \times 5.298}{0.0001 \cdot p} \approx \frac{158{,}940}{p}.$$

$p = I$	Required R for relative error ($\varepsilon = \delta = 0.01$)
0.5	$\approx 318{,}000$
0.1	$\approx 1.59 \times 10^6$
0.01	$\approx 1.59 \times 10^7$
0.001	$\approx 1.59 \times 10^8$
10^{-6}	$\approx 1.59 \times 10^{11}$

4.4.2 Absolute vs Relative Error: A Striking Contrast V

The contrast between the two tables is dramatic. For absolute error control with $p = 0.001$: only 7 replications needed. For relative error control with $p = 0.001$: 1.59×10^8 replications needed — an increase of ~ 23 million times.

The Fundamental Hardness of Rare-Event Simulation

For relative error control of a Bernoulli(p) estimator: required $R \propto 1/p$. Halving p doubles the required sample size. As $p \to 0$ (event becomes rarer), $R \to \infty$ proportionally.

At $p = 10^{-6}$: even if each simulation takes 1 nanosecond, 10^{11} replications would take 10^{11} nanoseconds $= 100$ seconds. At $p = 10^{-9}$, this becomes 10^{14} ns $=$ over a day. For very rare events (e.g., $p = 10^{-12}$ for catastrophic failure probabilities in aerospace engineering), naive Monte Carlo is completely impractical.

The solution is **variance reduction** (Chapter 5). Techniques such as **importance sampling** replace the original distribution with one that makes the rare event common, then correct for the change. An efficient importance sampling estimator can achieve relative error control with R independent of p — breaking the $1/p$ bottleneck.

4.4.3 Absolute vs Relative Error for Estimating π (pi) I

As a concrete bridge between the theory and practice, we now work through the complete sample-size calculation for estimating $I = \pi$ (pi, the mathematical constant ≈ 3.14159). Since π is of order 1 (it is neither tiny nor huge), we expect both error criteria to give manageable sample sizes — but relative error will still require somewhat more replications.

We use the **hit-or-miss estimator** from Example 4.1.6: $Y_j = 4 \cdot \mathbf{1}(U_{2j-1}^2 + U_{2j}^2 \leq 1)$, where U_{2j-1} and U_{2j} are i.i.d. $\mathcal{U}[0,1]$. Here $\mathbf{1}(\cdot)$ (indicator function) equals 1 if the point (U_{2j-1}, U_{2j}) lands inside the unit quarter-circle, 0 otherwise. This gives $\mathbb{E}Y = \pi$ and $\mathrm{Var}(Y) = 4\pi(4-\pi) \approx 2.697$.

We fix $\varepsilon = \delta = 0.01$ (1% accuracy, 99% confidence throughout).

4.4.3 Absolute vs Relative Error for Estimating π (pi) II

Sample Size for Absolute Error via CLT (Eq. 1.7)

We want $\mathbb{P}(|\hat{Y}_R - \pi| \geq \varepsilon) \leq \delta$ with $\varepsilon = 0.01$, i.e., the estimate is within ± 0.01 of π. The CLT formula gives:

$$R^{\text{abs}} = z_{0.995}^2 \cdot \frac{\text{Var}(Y)}{\varepsilon^2} = 2.576^2 \times \frac{2.697}{(0.01)^2} \approx 6.636 \times 26{,}970 \approx 1.79 \times 10^5.$$

So approximately **179,000 replications** are needed to guarantee that the estimate is within 0.01 of π with 99% confidence.

4.4.3 Absolute vs Relative Error for Estimating π (pi) III

Sample Size for Relative Error via Chernoff (Eq. 4.4)

We want $\mathbb{P}(|\hat{Y}_R - \pi|/\pi \geq \varepsilon) \leq \delta$ with $\varepsilon = 0.01$. Using the indicator $Y_j' = \mathbf{1}(U_{2j-1}^2 + U_{2j}^2 \leq 1)$ (which has $\mathbb{E}Y_j' = \pi/4 \approx 0.7854$), the Chernoff bound (4.4) with $p = \pi/4$ gives:

$$R^{\mathrm{rel},1} \geq \frac{3\ln(2/0.01)}{(0.01)^2 \cdot (\pi/4)} = \frac{3 \times 5.298}{0.0001 \times 0.7854} \approx \frac{15.894}{0.0000785} \approx 2.03 \times 10^5.$$

The estimate of $\pi/4$ must be within $\pm 1\%$ of its true value.

4.4.3 Absolute vs Relative Error for Estimating π (pi) IV

Sample Size for Relative Error via CLT

Equivalently, we require $|\hat{Y}_R - \pi| \leq \varepsilon\pi = 0.01\pi \approx 0.03142$ (making the absolute tolerance match 1% relative). Using the CLT formula with this tighter absolute tolerance:

$$R^{\text{rel},2} = z_{0.995}^2 \cdot \frac{\text{Var}(Y)}{(\varepsilon\pi)^2} = 2.576^2 \times \frac{2.697}{(0.01)^2 \times \pi^2} \approx \frac{1.79 \times 10^5}{9.870} \approx 1.81 \times 10^4.$$

Wait — the Chernoff-based bound (2.03×10^5) is *larger* than the CLT-based bound (1.81×10^4)? Yes: Chernoff is valid for any R but is conservative; the CLT is sharper (it accounts for the Gaussian shape of the distribution) but is only asymptotically valid. For $\pi \approx 3.14$, both bounds are manageable.

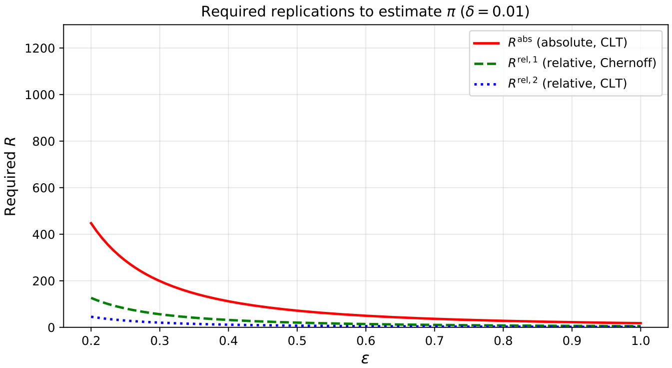
4.4.3 Absolute vs Relative Error for Estimating π (pi) V

Comparison: π vs a Rare Event with $p = 10^{-6}$

For π: relative error requires $\sim 2\times$ more replications than absolute error. For a rare event with $p = 10^{-6}$: absolute error requires only a handful of replications (trivially satisfied), while relative error requires $\sim 1.6 \times 10^{11}$ replications. The contrast is staggering.

The reason for the stark difference: $\pi \approx 3.14$ is not small, so the relative-error condition $|\hat{I}/I - 1| \leq \varepsilon$ does not demand much more than the absolute-error condition $|\hat{I} - I| \leq \varepsilon$. But for $p = 10^{-6}$, requiring $|\hat{I}/p - 1| \leq 0.01$ means the estimate must be accurate to within 10^{-8} in absolute terms — far harder than the absolute-error condition $|\hat{I} - p| \leq 0.01$ (which is trivially satisfied by estimating 0 every time).

4.4.3 Absolute vs Relative Error for Estimating π (pi) VI



4.4.4 How Variance Influences Both Error Criteria I

We now place the two error criteria side-by-side in a unified framework, using the CLT approximation throughout. For large R, the estimator $\hat{I} = \hat{Y}_R$ is approximately normally distributed: $\hat{I} \approx \mathcal{N}(I, \sigma_Y^2/R)$, where $\sigma_Y^2 = \mathrm{Var}(Y)$ (sigma sub Y squared, the variance of a single simulation output). This approximation captures both error criteria.

Required Replications: Unified Comparison

Error criterion	Condition	Required R
Absolute	$\mathbb{P}(\lvert\hat{I} - I\rvert \geq \varepsilon) \leq \delta$	$R = z_{1-\delta/2}^2 \dfrac{\sigma_Y^2}{\varepsilon^2}$
Relative	$\mathbb{P}(\lvert\hat{I}/I - 1\rvert \geq \varepsilon) \leq \delta$	$R = z_{1-\delta/2}^2 \dfrac{\sigma_Y^2}{\varepsilon^2 I^2}$

Here ε (epsilon) is the error tolerance, δ (delta) is the failure probability, $z_{1-\delta/2}$ (z sub one-minus-delta-over-two) is the standard normal quantile, σ_Y^2 is the single-sample variance, and I (capital I) is the true quantity being estimated.

4.4.4 How Variance Influences Both Error Criteria II

Notation

σ_Y^2 (sigma sub Y squared) — the variance of a single simulation output; this is the key "noise level" of the problem. Reducing σ_Y^2 directly reduces the required R.

$\varepsilon^2 I^2$ (epsilon squared times I squared) — the denominator for relative error; it includes the extra factor I^2 (I squared), which becomes tiny when I is small, dramatically increasing the required R.

The ratio of the two required sample sizes is: $R_{\mathrm{rel}}/R_{\mathrm{abs}} = 1/I^2$, which grows as $I \to 0$.

4.4.4 How Variance Influences Both Error Criteria III

How to Read This: The Role of Variance Reduction

Reducing the variance σ_Y^2 by a factor of c reduces the required R by the same factor c under *both* error criteria. This is why variance reduction techniques (importance sampling, stratified sampling, control variates, antithetic variates — all developed in Chapter 5) are valuable: they change the estimator from Y to a lower-variance Y^*, keeping the mean $\mathbb{E}Y^* = I$ the same but shrinking $\sigma_{Y^*}^2$.

For relative error with small I:

- Naive CMC (Crude Monte Carlo) requires $R \sim \sigma_Y^2/(\varepsilon^2 I^2)$; since $\sigma_Y^2 \geq I(1-I) \approx I$ for Bernoulli variables, this gives $R \gtrsim 1/(\varepsilon^2 I)$, growing unboundedly as $I \to 0$.
- A good importance sampling estimator can achieve $\sigma_{Y^*}^2 \propto I^2$, giving $R \sim 1/\varepsilon^2$ — independent of I. This is called **bounded relative error**: even as $I \to 0$, a fixed number of replications suffices.

Without variance reduction, naive Monte Carlo is fundamentally unsuited for rare events. With good variance reduction, the same framework applies.

Worked Example: Comparing Estimators for $I = 0.001$

Suppose $I = 0.001$ (the probability of a rare defect in manufacturing). We want 1% relative accuracy ($\varepsilon = 0.01$) with 99% confidence ($\delta = 0.01$, so $z_{0.995} = 2.576$).
Naive Bernoulli estimator: $Y_i \in \{0, 1\}$, $\sigma_Y^2 = I(1 - I) \approx 0.001$.

$$R_{\text{naive}} = 2.576^2 \times \frac{0.001}{(0.01)^2 \times (0.001)^2} \approx 6.636 \times 10^{10}.$$

Over 66 billion replications — completely impractical.
Perfect importance sampling: $\sigma_{Y^*}^2 = c \cdot I^2 = c \times 10^{-6}$ for some constant c (say $c = 1$ for a good estimator).

$$R_{\text{IS}} = 2.576^2 \times \frac{I^2}{(0.01)^2 \times I^2} = 2.576^2 \times \frac{1}{(0.01)^2} = 2.576^2 \times 10{,}000 \approx 66{,}355.$$

Only about 66,000 replications — a factor of 10^6 improvement.

4.4 Python: Comparing Bounds and Sample Size Calculations

```python
import numpy as np
from scipy import stats

# ---- Sample size formulas -----------------------------------------
def R_abs_clt(sigma2, epsilon, delta):
    """
    Required R for ABSOLUTE error: P(|Y_bar - I| >= epsilon) <= delta.
    Uses the CLT formula: R = z^2 * sigma^2 / epsilon^2.
    sigma2  : Var(Y), the single-sample variance
    epsilon : absolute error tolerance
    delta   : failure probability (e.g. 0.01 for 99% confidence)
    """
    z = stats.norm.ppf(1 - delta / 2)   # z_{1 - delta/2}
    return int(np.ceil(z**2 * sigma2 / epsilon**2))

def R_rel_clt(sigma2, I, epsilon, delta):
    """
    Required R for RELATIVE error: P(|Y_bar/I - 1| >= epsilon) <= delta.
    Uses CLT: R = z^2 * sigma^2 / (epsilon^2 * I^2).
    I : the true value being estimated
    """
    z = stats.norm.ppf(1 - delta / 2)
    return int(np.ceil(z**2 * sigma2 / (epsilon**2 * I**2)))

def R_rel_chernoff(p, epsilon, delta):
    """
    Required R for relative error, Chernoff bound (Bernoulli case).
```

4.5 Conclusions: The Fundamental Relationship Revisited I

Chapter 4 has developed a complete toolkit for quantifying the accuracy of Monte Carlo estimates based on independent replications. Every result in this chapter, regardless of what quantity is being estimated (mean, transformed mean, or quantile), leads back to a single unifying equation.

4.5 Conclusions: The Fundamental Relationship Revisited II

The Fundamental Relationship (Eq. 5.1)

$$b = \frac{z_{1-\alpha/2}\, D}{\sqrt{R}},$$

where:

- b (b) is the **confidence interval half-width**: the radius of the symmetric confidence interval centred at the point estimate. Smaller b means higher precision.
- R (capital R) is the **number of independent replications** (simulations) we run. The error shrinks as $1/\sqrt{R}$.
- $z_{1-\alpha/2}$ (z sub one-minus-alpha-over-two) is the **standard normal quantile** at level $1 - \alpha/2$. For 95% confidence: $\alpha = 0.05$, $z_{0.975} = 1.96$. For 99% confidence: $\alpha = 0.01$, $z_{0.995} = 2.576$.
- D (capital D) is the **problem's variability constant**, measuring the inherent difficulty of the estimation problem (unrelated to how many replications we run). See below for its form in each case.

4.5 Conclusions: The Fundamental Relationship Revisited III

The Variability Constant D for Each Estimation Problem

Quantity being estimated	Variability constant D	Section
$I = \mathbb{E}Y$ (1D mean)	$D = \sigma_Y = \sqrt{\mathrm{Var}(Y)}$	4.1
$k(I)$, $I = \mathbb{E}Y$ (transformed mean)	$D = \lvert k'(I)\rvert\, \sigma_Y$	4.2
$k(\mathbf{I})$, $\mathbf{I} = \mathbb{E}\mathbf{Y}$ (d-dim. transformed mean)	$D = \sqrt{\nabla k(\mathbf{I})\ \boldsymbol{\Sigma}_{\mathbf{Y}}\ (\nabla k(\mathbf{I}))^T}$	4.2
q_p (the p-th quantile)	$D = \sqrt{p(1-p)}\ /\ f(q_p)$	4.3

Here $\sigma_Y = \sqrt{\mathrm{Var}(Y)}$ (sigma sub Y) is the standard deviation of a single output; $k'(I)$ (k prime of I) is the derivative of the transformation; $\nabla k(\mathbf{I})$ (nabla k of I) is the gradient vector; $\boldsymbol{\Sigma}_{\mathbf{Y}}$ (bold Sigma sub Y) is the covariance matrix; $f(q_p)$ (f of q sub p) is the density at the quantile; and $p(1-p)$ (p times one-minus-p) is the binomial variance factor.

4.5 Conclusions: The Fundamental Relationship Revisited IV

How to Read the Fundamental Relationship

Rearranging $b = z_{1-\alpha/2} D/\sqrt{R}$ gives two equivalent forms:

$$R = z_{1-\alpha/2}^2 \frac{D^2}{b^2} \qquad \text{and} \qquad D = \frac{b\sqrt{R}}{z_{1-\alpha/2}}.$$

The first form answers: "How many replications do I need?" The second answers: "What variability level can I tolerate for a given budget?"

The crucial property is that b shrinks as $1/\sqrt{R}$. This has two important implications:

1. **Diminishing returns:** To halve the error b, you must *quadruple* R. There is no escaping this — it is a mathematical law, not a limitation of any particular algorithm.
2. **No curse of dimensionality:** The rate $b = O(R^{-1/2})$ holds regardless of the dimension d of the problem (whether $\mathbf{I}$ is 1-dimensional or 1000-dimensional). Deterministic integration methods (such as the trapezoidal rule) achieve $O(R^{-2/d})$ — exponentially slower in high dimensions. Monte Carlo's $O(R^{-1/2})$ rate is the reason it dominates in high-dimensional integration.

Practical workflow for a Monte Carlo study. The fundamental relationship guides a systematic four-step approach to any Monte Carlo estimation task.

1. **Define the estimator.** Express the quantity of interest as $I = \mathbb{E}Y$, or as $k(I)$ or q_p, and identify the corresponding variability constant D (using the table above).
2. **Pilot simulation.** Run a small "pilot" batch of m replications (typically $m = 200$ to $1{,}000$) to estimate D from the data. For a 1D mean: estimate σ_Y by the sample standard deviation $\hat{S}_m$. For a transformed mean: additionally compute $|k'(\hat{Y}_m)|$. For a quantile: additionally compute $\hat{Q}'_{p,m}$ via order statistics.
3. **Compute required R.** Plug the pilot estimates into $R = \lceil z^2_{1-\alpha/2}\hat{D}^2/b^2 \rceil$ (ceiling function, rounded up to the next integer).
4. **Main simulation and report.** Run R replications; compute the point estimate and its confidence interval using the formulas in the summary block below. Report both.

Summary of Confidence Intervals at 95% Confidence ($\alpha = 0.05$, $z_{0.975} = 1.96$)

Quantity	Point Estimate	95% Confidence Interval
$I = \mathbb{E}Y$	$\hat{Y}_R$	$\hat{Y}_R \pm \dfrac{1.96\,\hat{S}_R}{\sqrt{R}}$
$k(I)$, $I = \mathbb{E}Y$	$k(\hat{Y}_R)$	$k(\hat{Y}_R) \pm \dfrac{1.96\,\hat{S}_R\,\lvert k'(\hat{Y}_R)\rvert}{\sqrt{R}}$
$k(\mathbf{I})$, $\mathbf{I} = \mathbb{E}\mathbf{Y}$	$k(\hat{\mathbf{Y}}_R)$	$k(\hat{\mathbf{Y}}_R) \pm 1.96\,\dfrac{\hat{\varsigma}}{\sqrt{R}}$
q_p (quantile)	$X_{(\lfloor pR \rfloor)}$	$\hat{q}_p \pm 1.96\sqrt{\dfrac{p(1-p)}{R}}\,\hat{Q}'_{p,R}$

In the third row, $\hat{\varsigma}^2 = \sum_{i,j} k^{(i)}(\hat{\mathbf{Y}}_R)\,k^{(j)}(\hat{\mathbf{Y}}_R)\,\hat{\mathbf{S}}^2_R(i,j)$ is the estimated asymptotic variance from the multi-D delta method (Eq. 2.16).

4.5 Python: Complete Monte Carlo Output Analysis Workflow

```python
import numpy as np
from scipy import stats

def full_mc_analysis(simulate_Y, R_pilot=500, b=0.05, alpha=0.05, seed=42):
    """
    Complete four-step Monte Carlo output analysis pipeline.

    Parameters
    ----------
    simulate_Y : callable, takes an np.random.Generator and returns one float Y
    R_pilot    : int, number of pilot replications for variance estimation
    b          : float, desired CI half-width
    alpha      : float, significance level (e.g. 0.05 for 95% CI)
    seed       : int, random seed for reproducibility

    Returns
    -------
    Y_bar : float -- point estimate of I = E[Y]
    ci    : tuple (lo, hi) -- confidence interval
    """
    rng = np.random.default_rng(seed)

    # --- Step 1: Pilot simulation to estimate sigma^2 = Var(Y) ---------------
    Y_pilot     = np.array([simulate_Y(rng) for _ in range(R_pilot)])
    sigma2_hat = Y_pilot.var(ddof=1)     # unbiased estimate: divide by R_pilot - 1
    print(f"Pilot sigma^2 = {sigma2_hat:.4f}  (from {R_pilot} pilot runs)")
```

4.5 Key Takeaways of Chapter 4 I

This frame collects the six central lessons of Chapter 4. Understanding these deeply provides the foundation for all subsequent variance reduction and advanced simulation methodology.

1. The Monte Carlo Rate Is Universal and Dimension-Free

The confidence interval half-width satisfies $b = z_{1-\alpha/2} D/\sqrt{R}$, so b decreases as $O(R^{-1/2})$ regardless of the dimension d of the problem. This is the decisive advantage of Monte Carlo over deterministic numerical integration, which suffers the *curse of dimensionality* (the number of function evaluations needed for fixed accuracy grows exponentially with d for grid-based methods). For high-dimensional problems ($d \geq 5$), Monte Carlo is often the only practical approach.

2. CLT Is Much Sharper Than Chebyshev

For 95% confidence, the CLT-based confidence interval has half-width $z_{0.975}\sigma_Y/\sqrt{R} = 1.96\sigma_Y/\sqrt{R}$, while the Chebyshev bound gives $\sigma_Y/(\sqrt{0.05}\sqrt{R}) \approx 4.47\sigma_Y/\sqrt{R}$. The ratio is $1.96/4.47 \approx 0.438$, meaning the CLT-based required sample size is $(0.438)^2 \approx 19\%$ of the Chebyshev required sample size — a $\approx 5\times$ reduction. The price is that the CLT holds only asymptotically (for large R), whereas Chebyshev is exact for all R. In practice, the CLT approximation is excellent for $R \geq 30$ and the distribution of Y is not extremely heavy-tailed.

3. The Delta Method for Transformed Estimators

When the quantity of interest is $z = k(I)$ rather than I itself, the natural estimator $k(\hat{Y}_R)$ is generally biased. The bias is of order $O(R^{-1})$ and equals $\beta^Z = \sigma_Y^2 k''(I)/2$ (when $\hat{Y}_R$ is unbiased), from Jensen's inequality. However, the standard deviation of $k(\hat{Y}_R)$ is of order $O(R^{-1/2})$, which dominates the bias for large R. Therefore bias corrections are typically unnecessary, and the delta method CI (Eq. 2.3) has the correct asymptotic coverage.

The delta method replaces the asymptotic variance σ_Y^2 by $k'(I)^2\sigma_Y^2$: the transformation amplifies uncertainty where k is steep ($|k'(I)|$ large) and compresses it where k is flat ($|k'(I)|$ small).

4. Quantile Estimation: Density Matters

Estimating the p-th quantile q_p is harder than estimating a mean. The asymptotic variance is $\varsigma^2 = p(1-p)/f(q_p)^2$, which grows large when the density $f(q_p)$ is small (quantile is in a sparse region of the distribution). The density must be estimated from the data using an order-statistic finite-difference formula with bandwidth $m_R \sim \sqrt{R}$ (simple) or $m_R \sim R^{4/5}$ (MSE-optimal). The resulting confidence interval converges at the usual Monte Carlo rate $O(R^{-1/2})$, but its asymptotic variance constant ς^2 is typically larger than for mean estimation.

5. Relative Error Is Exponentially Harder Than Absolute Error for Rare Events

For a Bernoulli(p) estimator:

- Absolute error ($|\hat{Y}_R - p| \le \varepsilon$): requires $R \sim p(1-p)/\varepsilon^2$, which *decreases* as $p \to 0$. This is misleading — a trivial estimate of 0 satisfies the criterion.
- Relative error ($|\hat{Y}_R/p - 1| \le \varepsilon$): requires $R \sim 1/(\varepsilon^2 p)$, which *increases* as $p \to 0$. For $p = 10^{-6}$ and $\varepsilon = 0.01$: over 10^{11} replications needed — completely infeasible for naive Monte Carlo.

Relative error is the correct criterion for rare events. Satisfying it with naive Monte Carlo is impossible for very small p; this *motivates all of Chapter 5* on variance reduction, particularly importance sampling.

6. Multivariate Confidence Regions Have Three Shapes

When estimating a vector mean $\mathbf{I} = \mathbb{E}\mathbf{Y} \in \mathbb{R}^d$:

- **Ellipsoid** (Eq. 1.15): the smallest region achieving coverage $1-\alpha$ exactly; requires the full covariance matrix $\mathbf{\Sigma}_{\mathbf{Y}}$ and its eigendecomposition.
- **Parallelogram** (Eq. 1.18): slightly larger, aligned with the eigenvectors of $\mathbf{\Sigma}_{\mathbf{Y}}$; also achieves exactly $1-\alpha$.
- **Rectangle** (Eq. 1.19): largest, axis-aligned (ignores correlations); coverage is at least $1-\alpha$ (conservative), but easiest to compute and communicate.

As the dimension d increases, all three regions grow in size because the chi-squared quantile $q_{1-\alpha}(\chi^2_d)$ increases with d, reflecting the increased difficulty of simultaneously estimating more components.

Chapter 4: Visual Summary

Section 4.1 — Estimating $I = \mathbb{E}Y$

- $\hat{Y}_R = \frac{1}{R}\sum Y_j$: unbiased, strongly consistent
- Chebyshev: $b_R = \sigma_Y/(\sqrt{\alpha}\sqrt{R})$ (valid for any R, conservative)
- CLT: $b = z_{1-\alpha/2}\sigma_Y/\sqrt{R}$ (sharp, holds asymptotically)
- Required $R = z_{1-\alpha/2}^2\sigma_Y^2/b^2$
- CLT needs only $\approx 19\%$ of Chebyshev's R
- Online updates (Eqs. 1.11–1.12) for large-scale streaming
- Multivariate: confidence ellipsoid, parallelogram, rectangle

Section 4.2 — Estimating $k(I)$

- Delta method: $\varsigma^2 = k'(I)^2\sigma_Y^2$
- Multi-D: $\varsigma^2 = \nabla k\ \mathbf{\Sigma_Y}\ (\nabla k)^T$
- Bias $\sim \sigma_Y^2 k''(I)/(2R)$: order $O(R^{-1})$, dominated by statistical $O(R^{-1/2})$

Section 4.3 — Quantile Estimation

- $\hat{q}_p = X_{(\lfloor pR \rfloor)}$: consistent, asymptotically normal
- Asymptotic variance: $\varsigma^2 = p(1-p)/f(q_p)^2$
- Estimate $1/f(q_p)$ via order-statistic finite differences (Eq. 3.3)
- Bandwidth: $m_R \sim \sqrt{R}$ (simple) or $m_R \sim R^{4/5}$ (optimal)
- Tail quantiles harder: smaller $f(q_p) \Rightarrow$ larger ς^2

Section 4.4 — Error Bounds

- Chernoff: $\mathbb{P} \leq 2e^{-\varepsilon^2 Rp/3}$ (exponential, vs polynomial Chebyshev)
- Absolute error: $R \propto \sigma_Y^2/\varepsilon^2$ (decreases for rare events — misleading)
- Relative error: $R \propto \sigma_Y^2/(\varepsilon^2 I^2)$ (grows as $1/I$ for rare events — correct criterion)
- For $I = 10^{-6}$: need $\sim 10^{11}$ naive runs — variance reduction essential